MODULE - 04

ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER

* Linear differential equation of second and higher order with constant co-efficients.

Basically a D.E is said to be linear if the dependent variable and its derivatives are of first degree.

A DE of the form,

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1}\frac{dy}{dx} + a_n y = \phi(x)$$

where $a_1, a_2, \ldots a_n$ are all constants is called a linear differential equation of n^{th} order with constant coefficients.

The familiar notation of differential operator :

$$D = \frac{d}{dx},\quad D^2 = \frac{d^2}{dx^2},\quad \ldots\ D^n = \frac{d^n}{dx^n}$$

the differential equation can be put in the form,

$$[D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n]y = \phi(x)$$

i.e., $\boxed{f(D)\, y = \phi(x)}$ —— (1)

$a_0, a_1, a_2, \ldots a_n$ are arbitrary constants.

where $f(D)$ is a polynomial in D of degree n.

$f(D)$ = function of D

$\phi(x)$ = function of x.

∴ The General soln for eqn (1) is

If $\phi(x) = 0$ the equation $f(D)y = 0$ is called a homogeneous equation.

If $\phi(x) \neq 0$ then the equation (1) is called a non-homogeneous equation.

* Solution of homogeneous linear differential equation.

Consider the equation in the form

$$\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2 y = 0$$

where a_1, a_2 are constants — (1)

i.e $(D^2 + a_1 D + a_2)y = 0$ or $f(D)y = 0$ — (2)

where $f(D) = D^2 + a_1 D + a_2$.

Firstly we are going to establish the following fundamental property:

If y_1 & y_2 are two linearly independent solutions of (2) then $c_1y_1 + c_2y_2$ is also a soln of (2) where c_1 & c_2 are arbitrary constants.

Since y_1 & y_2 are solutions of (2) we have,

$$f(D)y_1 = 0 \quad \& \quad f(D)y_2 = 0$$

$$\therefore \quad f(D)[c_1y_1 + c_2y_2] = c_1 f(D)y_1 + c_2 f(D)y_2 = c_1 \cdot 0 + c_2 \cdot 0 = 0$$

thus $f(D)[c_1y_1 + c_2y_2] = 0 \Rightarrow c_1y_1 + c_2y_2$ is a soln of (2).

Since the general solⁿ of a second order differential equation has to contain two arbitrary constants $c_1y_1 + c_2y_2$ in the general solution of ②.

Denoting $y_c = c_1y_1 + c_2y_2$ we have $f(D)y_c = 0$

y_c is called as Complimentary Function (C.F) and it is the solution of homogeneous D.E ①.

* Solution of non-homogeneous linear differential equation.

Let us consider $f(D)y = \phi(x)$

Also, let $y = y_p(x)$ be a particular solution (without the involvement of arbitrary constants) of ①.

Hence we must have,

$$f(D)\, y_p = \phi(x)$$

Now, $f(D)[y_c + y_p] = f(D)y_c + f(D)y_p$

$= 0 + \phi(x) = \phi(x)$

i.e, $f(D)[y_c + y_p] = \phi(x)$

$\Rightarrow \quad y = y_c + y_p$

Thus y as given by ⑤ is also a solution of ① & is called the general solution or Complete solution of the given non-homogeneous equation.

The particular soln y_p is also called as Particular Integral P.I.

(4)

* **Note:**

The general solution of a non-homogeneous equation consists of two parts namely the complimentary function and the particular integral.

The general solution is of the form

$$y = C.F + P.I \quad (or) \quad y = y_c + y_p.$$

Obviously if $\phi(x) = 0$, the complementary function itself is the general solution.

* **Method of finding the complimentary function.**

consider a second order homogeneous DE put in the form,

$$f(D)y = 0.$$

where $f(D) = D^2 + a_1 D + a_2$

we form the 'Auxiliary Equation' (AE), $f(m) = 0$ & solve the same to obtain two roots m_1 & m_2 which can be in the following three forms,

(i) real & distinct

(ii) real & coincident

(iii) complex pair.

→ The complimentary function (CF) y_c for these three cases is as follows,

Case (i): $y = y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ $\quad (m_1 \neq m_2)$

also if $m_1, m_2, m_3 \dots m_n$ are the 'n' real mutually distinct roots of an n^{th} order equation, then,

$$y = y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

Case (ii): $m_1 = m_2 = m$

$$y = y_c = (c_1 + c_2 x) e^{mx}$$

also if $m_1 = m_2 = m_3 = \dots m_n = m$ (say) of an n^{th} order equation, then,

$$y = y_c = (c_1 + c_2 x + c_3 x^2 + \dots + c_n x^{n-1}) e^{mx}$$

Case (iii): m_1, m_2 are complex pair of roots say $p \pm iq$

$$y = y_c = e^{px} (c_1 \cos qx + c_2 \sin qx)$$

also if, $p \pm iq$, (say) are the repeated complex pair of roots then,

$$y = y_c = e^{px} \{(c_1 + c_2 x) \cos qx + (c_3 + c_4 x) \sin qx\}$$ & so on.

* Table for writing the C.F based on roots of A.E.

	Roots of AE	C.F
(1)	2, 3	$c_1 e^{2x} + c_2 e^{3x}$
(2)	1, −1, 2, −2	$c_1 e^{x} + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x}$
(3)	3, 3	$(c_1 + c_2 x) e^{3x}$
(4)	0, −2, −2, −2	$c_1 e^{0x} + (c_2 + c_3 x + c_4 x^2) e^{-2x}$
(5)	$\pm 2i$	$c_1 \cos 2x + c_2 \sin 2x$

6	$3 \pm 2i$	$e^{3x}(c_1\cos 2x + c_2\sin 2x)$
7	$1, 1, 1 \pm i$	$(c_1 + c_2 x)e^x + e^x(c_3\cos x + c_4\sin x)$
8	$1 \pm 2i, 1 \pm 2i$	$e^x\{(c_1 + c_2 x)\cos 2x + (c_3 + c_4 x)\sin 2x\}$

* Working Procedure for solving problems (solution of homogeneous DE).

→ Step ①:- The given DE is put in the form $f(D)y = 0$.

→ Step ②:- Form the AE, $f(m) = 0$ & solve the same. [Here we need to adopt various techniques to solve the equation including the synthetic division method].

→ Step ③:- Based on the nature of the roots of the AE we write the CF which itself is the general solution of the DE taking into account the dependent & the independent variables involved in DE.

* PROBLEMS:

① Solve $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 8y = 0$.

→ we have $(D^3 - 2D^2 + 4D - 8)y = 0$ where $D = \frac{d}{dx}$

A.E is $m^3 - 2m^2 + 4m - 8 = 0$ (factorization)

$m^2(m-2) + 4(m-2) = 0$

$(m-2)(m^2+4) = 0$

$\Rightarrow m = 2, \quad m = \sqrt{-4} = \sqrt{4i^2} = \pm 2i$

Roots of the AE are $2, \pm 2i$

Thus, $\boxed{y = c_1 e^{2x} + c_2 \cos 2x + c_3 \sin 2x}$ is the general solution.

(5) Solve $\dfrac{d^3y}{dx^3} + 6\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 0.$

$\rightarrow$ we have $(D^3 + 6D^2 + 11D + 6)y = 0.$

AE is, $m^3 + 6m^2 + 11m + 6 = 0$ (This cannot be factorized).

we shall find one root by inspection by taking values for $m = 1, -1, 2, -2$ etc.

Since all the terms are positive we have to try only negative values for m.

putting $m = -1$ we have

$(-1)^3 + 6(-1)^2 + 11(-1) + 6 = -1 + 6 - 11 + 6 = 0$

$\therefore m = -1$ is a root.

The other roots can be found through the process of synthetic division

$$\begin{array}{r|rrrr} -1 & 1 & 6 & 11 & 6 \\ & 0 & -1 & -5 & -6 \\ \hline & 1 & 5 & 6 & \boxed{0} \end{array}$$

$\longrightarrow$ all the coefficients are written

$\longrightarrow$ zero to be written below the 1st coefficient & added

$\longrightarrow$ resultant to be multiplied with -1 & added.

Since the last figure is zero we respectively attach m^2, m to the first two coefficients.

That is, $m^2+5m+6=0$.

$$(m+2)(m+3)=0$$
$$m=-2,-3$$

Hence $m=-1,-2,-3$ are the roots of A.E which are all real & distinct.

Thus $\boxed{y=c_1e^{-x}+c_2e^{-2x}+c_3e^{-3x}}$ is the general soln.

③ Solve : $(D^3-3D+2)y=0$.

→ A.E is $m^3-3m+2=0$

$m=1$ is a root by inspection.

The other two roots can be found through synthetic division.

$$\begin{array}{c|cccc} 1 & 1 & 0 & -3 & 2 \\ & 0 & 1 & 1 & -2 \\ \hline & 1 & 1 & -2 & \boxed{0} \end{array}$$

Now $m^2+m-2=0$ (or)

$$(m+2)(m-1)=0$$
$$\Rightarrow m=-2,1$$

Hence $m=1,1,-2$ are the roots of A.E

Thus, $\boxed{y=(c_1+c_2x)e^{x}+c_3e^{-2x}}$ is the general soln.

④ Solve $(4D^4-4D^3-23D^2+12D+36)y=0$

→ A.E is $4m^4-4m^3-23m^2+12m+36=0$

If $m=2$; $64-32-92+24+36=124-124=0$

$\Rightarrow$ $m=2$ is a root by inspection.

Now by synthetic division,

$$\begin{array}{c|ccccc} 2 & 4 & -4 & -23 & 12 & 36 \\ & 0 & 8 & 8 & -30 & -36 \\ \hline & 4 & 4 & -15 & -18 & \boxed{0} \end{array}$$

now $4m^3+4m^2-15m-18=0$

If $m=2$, $32+16-30-18 = 48-48 = 0$

Again $m=2$ is a root. By synthetic division.

$$\begin{array}{c|cccc} 2 & 4 & 4 & -15 & -18 \\ & 0 & 8 & 24 & 18 \\ \hline & 4 & 12 & 9 & 0 \end{array}$$

now $4m^2+12m+9=0$ (or)

$(2m+3)^2=0 \Rightarrow m=-3/2, -3/2$

Thus $\boxed{y=(c_1+c_2x)e^{2x}+(c_3+c_4x)e^{-3x/2}}$ is the general soln.

(5) solve : $\frac{d^3y}{dx^3}+y=0$

$\rightarrow$ we have $(D^3+1)y=0$

A.E is $m^3+1=0$, $m=-1$ is a root by inspection.

The other two roots are found by synthetic division.

$$\begin{array}{c|cccc} -1 & 1 & 0 & 0 & 1 \\ & 0 & -1 & 1 & -1 \\ \hline & 1 & -1 & 1 & 0 \end{array}$$

now let us solve $m^2-m+1=0$

$\therefore m = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{3}}{2} = \frac{1 \pm \sqrt{3i^2}}{2} = \frac{1 \pm i\sqrt{3}}{2}$

Hence $m = -1,\ \frac{1}{2} \pm i(\sqrt{3}/2)$ are the roots of A.E.

Thus $\boxed{y = c_1 e^{-x} + e^{x/2}\{c_2 \cos(\sqrt{3}/2)x + c_3 \sin(\sqrt{3}/2)x\}}$

(6) Solve $(D^4 - 5D^2 + 4)y = 0$

$\rightarrow$

A.E is $m^4 - 5m^2 + 4 = 0$ (or) $(m^2)^2 - 5m^2 + 4 = 0$

$(m^2 - 1)(m^2 - 4) = 0$

$(m-1)(m+1)(m-2)(m+2) = 0$

$\therefore$ roots of A.E are $1, -1, 2, -2$

Thus $\boxed{y = c_1 e^{x} + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x}}$ is the general soln.

(7) Solve : $\frac{d^4 y}{dt^4} + 8\frac{d^2 y}{dt^2} + 16y = 0$

$\rightarrow$

We have $(D^4 + 8D^2 + 16)y = 0$ where $D = \frac{d}{dt}$

A.E is, $m^4 + 8m^2 + 16 = 0$ (or) $(m^2 + 4)^2 = 0$

$\therefore$ roots of A.E are $\pm 2i, \pm 2i$ (repeated imaginary roots).

['t' is the independent variable]

Thus, $\boxed{y = (c_1 + c_2 t)\cos 2t + (c_3 + c_4 t)\sin 2t}$ is the general solution.

(8) Solve : $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$

→ A.E is $4m^4 - 8m^3 - 7m^2 + 11m + 6 = 0$

If $m = -1$; $4 + 8 - 7 - 11 + 6 = 18 - 18 = 0$

$\therefore m = -1$ is a root by inspection.

-1	4	-8	-7	11	6
	0	-4	12	-5	-6
	4	-12	5	6	0

now, $4m^3 - 12m^2 + 5m + 6 = 0$

If $m = 2$; $32 - 48 + 10 + 6 = 48 - 48 = 0$

$\therefore m = 2$ is also a root.

2	4	-12	5	6
	0	8	-8	-6
	4	-4	-3	0

now $4m^2 - 4m - 3 = 0$

$\therefore 4m^2 - 6m + 2m - 3 = 0$

$2m(2m - 3) + 1(2m - 3) = 0$

$(2m + 1)(2m - 3) = 0$

$\Rightarrow m = -1/2, 3/2$

Hence the roots of the AE are $-1, 2, -1/2, 3/2$

Thus $\boxed{y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{-x/2} + c_4 e^{3x/2}}$ is the general soln.

(9) $\frac{d^4x}{dt^4} = m^4 x$ S.T. $x = c_1 \cos mt + c_2 \sin mt + c_3 \cosh mt + c_4 \sinh mt$

→ we have $(D^4 - m^4)x = 0$ where $D = \frac{d}{dt}$

(note: Since the given example involves 'm' we should not use the same for writing the AE)

i.e $p = \pm mi$, $p = \pm m$ are the roots of AE, Also $x = x(t)$

$\therefore x = c_1 \cos mt + c_2 \sin mt + c_3 e^{mt} + c_4 e^{-mt}$ — (1)

consider the desired expression.

$$x = c_1 \cos mt + c_2 \sin mt + c_3 \cosh mt + c_4 \sinh mt$$

$$x = c_1 \cos mt + c_2 \sin mt + c_3\left[\frac{e^{mt}+e^{-mt}}{2}\right] + c_4\left[\frac{e^{mt}-e^{-mt}}{2}\right]$$

$$x = c_1 \cos mt + c_2 \sin mt + \left[\frac{c_3+c_4}{2}\right]e^{mt} + \left[\frac{c_3-c_4}{2}\right]e^{-mt}$$

Denoting $c_3' = \dfrac{c_3+c_4}{2}$ and $c_4' = \dfrac{c_3-c_4}{2}$ we get,

$$\boxed{x = c_1 \cos mt + c_2 \sin mt + c_3' e^{mt} + c_4' e^{-mt}} \quad \text{— (2)}$$

we conclude that (1) = (2) since c_3', c_4' are also arbitrary constants.

*** Solution of homogeneous D.E subject to given conditions.**

(10) solve $y'' + 4y' + 4y = 0$ given that $y = 0$, $y' = -1$ at $x = 1$

→ we have $(D^2 + 4D + 4)y = 0$ where $D = \dfrac{d}{dx}$

A.E is $m^2 + 4m + 4 = 0$ (or) $(m+2)^2 = 0$

$m = -2, -2$

Hence $y = (c_1 + c_2 x)e^{-2x}$ — (1)

now, $y' = (c_1 + c_2 x)(-2e^{-2x}) + c_2 e^{-2x}$ — (2)

consider $y = 0$ at $x = 1$. hence (1) becomes

$0 = (c_1 + c_2)e^{-2} \Rightarrow c_1 + c_2 = 0$

Also, $y' = -1$ at $x = 1$.

Hence ② becomes

$$-1 = (c_1+c_2)(-2e^{-2}) + c_2 e^{-2} \quad , \text{ But } c_1+c_2 = 0$$

ie $-1 = c_2(e^{-2})$ (or) $c_2 = -e^2$

$\therefore c_1 = -c_2 = e^2$

sub these values in ① we get

$$y = e^2(1-x)e^{-2x}$$

Thus, $\boxed{y = (1-x)e^{2(1-x)}}$ is the particular solution.

(11) Solve : $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 5y = 0$, given that $y = 2$ & $\frac{dy}{dx} = \frac{d^2y}{dx^2}$ when $x = 0$.

$\rightarrow$ we have $(D^2 + 4D + 5)y = 0$

A.E $m^2 + 4m + 5 = 0$

$$\therefore m = \frac{-4 \pm \sqrt{16-20}}{2} = \frac{-4 \pm 2i}{2}$$

$$m = -2 \pm i$$

Hence, $y = e^{-2x}(c_1\cos x + c_2\sin x) \longrightarrow$ ①

$$\frac{dy}{dx} = y' = e^{-2x}(-c_1\sin x + c_2\cos x) - 2e^{-2x}(c_1\cos x + c_2\sin x)$$

ie, $y' = e^{-2x}(-c_1\sin x + c_2\cos x) - 2y$

$$y'' = e^{-2x}(-c_1\cos x - c_2\sin x) - 2e^{-2x}(-c_1\sin x + c_2\cos x) - 2y'$$

Consider, $y = 2$ when $x = 0$.

$y|_{x=0} = c_1$ from ①.

$\therefore \boxed{2 = c_1}$

Consider, $y' = y''$ when $x = 0$

$$y'|_{x=0} = c_2 - 2c_1$$

$$y''|_{x=0} = -c_1 - 2c_2 - 2(c_2 - 2c_1)$$

$$\therefore \quad c_2 - 2c_1 = 3c_1 - 4c_2 \text{ (or)}$$

$$5c_1 - 5c_2 = 0$$

$$\Rightarrow \boxed{c_1 = c_2}$$

But $c_1 = 2 \quad \therefore c_2 = 2$

Thus by substituting $c_1 = 2 = c_2$ in ① we obtain the particular solution.

$$\boxed{y = 2e^{-2x}(\cos x + \sin x)}$$ //.

* Inverse Differential Operator & the Particular Integral

we have the differential operator $D = \frac{d}{dx}$

Let us suppose that $D[G(x)] = F(x)$

Then $\int F(x)\,dx = G(x)$ is an inverse operation.

Equivalently we write $\frac{1}{D} F(x)\,dx = G(x)$

The operator $\frac{1}{D}$ that stands for the integral is called an inverse differential operator.

$\frac{1}{D^2}, \frac{1}{D^3}$ etc, stands for successive integration.

Example: $\frac{1}{D} x^2 = \int x^2\,dx = \frac{x^3}{3}$, $\frac{1}{D} \cos x = \sin x$.

& $\frac{1}{D^2} e^{3x} = \iint e^{3x}\,dx\,dx = \int \frac{e^{3x}}{3}\,dx = \frac{e^{3x}}{9}$.

→ next, if 'a' is a constant let us find $\frac{1}{D-a} F(x)$

Suppose $\frac{1}{D-a} F(x) = y$ then

$$(D-a)\,y = F(x) \quad (\text{or}) \quad \frac{dy}{dx} - ay = F(x)$$

which is a linear differential equation whose soln is given by

$$y\,e^{-ax} = \int F(x)\,e^{-ax}\,dx \quad (\text{or}) \quad y = e^{ax}\int F(x)\,e^{-ax}\,dx$$

$$\frac{1}{D-a} F(x) = e^{ax}\int F(x)\,e^{-ax}\,dx \quad \text{—— (1)}$$

In general we can say that,

$$f(D)y = \phi(x).$$

$$\boxed{y = \frac{\phi(x)}{f(D)}}$$

But we have already said that if $f(D)y = \phi(x)$ then $y = y_p(x)$ free from the arbitrary constants is the particular sol^n (or) particular integral (P.I)

$$\boxed{P.I = y_p = \frac{\phi(x)}{f(D)}}$$

Further it may also be noted that if $\phi(x) = a\phi_1(x) + b\phi_2(x)$, where a & b are constants, then.

$$P.I = y_p = \frac{\phi(x)}{f(D)} = a\frac{\phi_1(x)}{f(D)} + b\frac{\phi_2(x)}{f(D)}$$

Example: Finding P.I by the basic method

① Let us find $\frac{1}{D-1}\cos 2x$

→ using $\frac{1}{D-a}F(x) = e^{ax}\int e^{-ax}F(x)\,dx$. we have

$$\frac{1}{D-1}\cos 2x = e^{x}\int e^{-x}\cos 2x\,dx$$

Using the standard integral,

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$$ in RHS ① we get

$$\frac{1}{D-1}\cos 2x = e^{x}\cdot\frac{e^{-x}}{5}(-\cos 2x + 2\sin 2x)$$

Hence, $\frac{1}{D-1}\cos 2x = \frac{1}{5}(2\sin 2x - \cos 2x)$

(5) Let us find $\frac{1}{D^2+3D+2}e^x$

→ here $\frac{1}{D^2+3D+2} = \frac{1}{(D+1)(D+2)}$

$$= \frac{1}{D+1} - \frac{1}{D+2} \quad \text{by partial } f^n.$$

$$\therefore \frac{1}{D^2+3D+2}e^x = \frac{1}{D+1}e^x - \frac{1}{D+2}e^x$$

$$= e^{-x}\int e^x . e^x dx - e^{-2x}\int e^{2x}.e^x dx, \quad \text{by using (1).}$$

$\frac{1}{D-a}f(x) = e^{ax}\int f(x)e^{-ax}dx$

$$\frac{1}{D^2+3D+2}e^x = e^{-x}.\frac{e^{2x}}{2} - e^{-2x}.\frac{e^{3x}}{3}$$

$$= \frac{e^x}{2} - \frac{e^x}{3}$$

$$\boxed{\frac{1}{D^2+3D+2}.e^x = \frac{e^x}{6}}$$ //.

This fundamental method becomes highly tedious when $f(D)$ is in higher orders & $\phi(x)$ is in a complicated form as we have to necessarily decompose $f(D)$ into the form $(D-m_1)(D-m_2)---(D-m_n)$ for applying the result.

Hence we proceed to illustrate simple & shortcut established methods for finding the PI when $\phi(x)$ in $f(D)y=\phi(x)$ belong to some specific form.

* We discuss five specific forms of particular integral.

→ **TYPE - 01** : P.I of the form $\frac{e^{ax}}{f(D)}$

$$\frac{e^{ax}}{f(D)} = \frac{e^{ax}}{f(a)} \quad (D \text{ being replaced by } a)$$

where $f(a) \neq 0$.

$$\frac{e^{ax}}{f(D)} = x \cdot \frac{e^{ax}}{f'(a)} \quad \text{where } f(a) = 0 \ \& \ f'(a) \neq 0.$$

This result can be extended further also.

i.e. if $f'(a) = 0$, $\frac{e^{ax}}{f(D)} = x^2 \cdot \frac{e^{ax}}{f''(a)}$ & so on.

* note :- The result also holds good for the following functions.

(i) e^{ax+b} (ii) any constant k as we can write $k = k \cdot e^{0x}$

(iii) $a^x = (e^{\log a})^x = e^{bx}$ where $b = \log a$

(iv) $\sinh ax = \frac{1}{2}(e^{ax} - e^{-ax})$ & $\cosh ax = \frac{1}{2}(e^{ax} + e^{-ax})$.

ex :- (1) $\frac{e^x}{D^2+3D+2} = \frac{e^x}{1^2+3 \cdot 1+2} = \frac{e^x}{6}$

(2) $\frac{e^{3x+4}}{D^3+D^2+D+1} = \frac{e^{3x+4}}{(3)^3+(3)^2+3+1} = \frac{e^{3x+4}}{40}$

\# **Problems**

Type - 1

① Solve $6\frac{d^2y}{dx^2} + 17\frac{dy}{dx} + 12y = e^{-x}$

→ We have $(6D^2 + 17D + 12)y = e^{-x}$

A.E is, $6m^2 + 17m + 12 = 0$

$$6m^2 + 9m + 8m + 12 = 0$$

$$3m(2m+3) + 4(2m+3) = 0$$

$$(2m+3)(3m+4) = 0$$

$$m = -3/2, -4/3$$

$\therefore \; y_c = c_1 e^{-3x/2} + c_2 e^{-4x/3}.$

$$y_p = \frac{e^{-x}}{6D^2+17D+12} = \frac{e^{-x}}{6(-1)^2+17(-1)+12} = \frac{e^{-x}}{1} = e^{-x}$$

Complete solution: $y = y_c + y_p$ or $y = C.F + P.I$

Thus, $\boxed{y = c_1 e^{-3x/2} + c_2 e^{-4x/3} + e^{-x}}$

② Solve : $y'' + 2y' + y = \cosh(x/2)$

→ $(D^2 + 2D + 1)y = \cosh(x/2)$

A.E is $m^2 + 2m + 1 = 0$ (or) $(m+1)^2 = 0$

$\Rightarrow m = -1, -1$

$\therefore \; y_c = (c_1 + c_2 x)e^{-x}$

$$y_p = \frac{1}{2} \cdot \frac{e^{x/2} + e^{-x/2}}{(D+1)^2} = \frac{1}{2}\left[\frac{e^{x/2}}{(D+1)^2} + \frac{e^{-x/2}}{(D+1)^2}\right]$$

$$y_p = \frac{1}{2}\left[\frac{e^{x/2}}{(1/2+1)^2} + \frac{e^{-x/2}}{(-1/2+1)^2}\right]$$

$$y_p = \frac{1}{2} \cdot \frac{4}{9} e^{x/2} + \frac{1}{2} \cdot \frac{e^{-x/2}}{(1/4)} = \frac{2}{9} e^{x/2} + 2e^{-x/2}$$

$$P.I = y_p = \frac{2}{9} e^{x/2} + 2e^{-x/2}$$

complete solution $y = y_c + y_p$

Thus, $\boxed{y = (c_1 + c_2 x)e^{-x} + 2/9 \cdot e^{x/2} + 2e^{-x/2}}$ //.

③ Solve : $(D^3 - D^2 + 4D - 4)y = \sinh(2x+3)$

$\rightarrow$ AE is $m^3 - m^2 + 4m - 4 = 0$

$m^2(m-1) + 4(m-1) = 0$

$(m-1)(m^2+4) = 0 \Rightarrow m = 1, \; m^2 = -4$

$m = \pm 2i$

$\Rightarrow m = 1, \pm 2i$

$\therefore y_c = c_1 e^x + c_2 \cos 2x + c_3 \sin 2x$

$$y_p = \frac{\sinh(2x+3)}{D^3 - D^2 + 4D - 4} = \frac{1}{2}\left[\frac{e^{2x+3}}{D^3 - D^2 + 4D - 4} - \frac{e^{-(2x+3)}}{D^3 - D^2 + 4D - 4}\right]$$

$$y_p = \frac{1}{2}\left[\frac{e^{2x+3}}{8-4+8-4} - \frac{e^{-(2x+3)}}{-8-4-8-4}\right]$$

$$= \frac{1}{2}\left[\frac{e^{2x+3}}{8} - \frac{e^{-(2x+3)}}{-24}\right]$$

$$y_p = \frac{1}{48}\left[3e^{2x+3} + e^{-(2x+3)}\right]$$

Complete soln $y = y_c + y_p$

Thus, $y = c_1 e^{x} + c_2 \cos 2x + c_3 \sin 2x + \frac{1}{48}\left[3e^{2x+3} + e^{-(2x+3)}\right]$

(4) Solve : $\frac{d^2y}{dx^2} - 4y = \cosh(2x-1) + 3^x$

$\rightarrow$ $(D^2 - 4)y = \cosh(2x-1) + 3^x$

A.E is $m^2 - 4 = 0$

$(m-2)(m+2) = 0$

$m = 2, -2$

$C.F = y_c = c_1 e^{2x} + c_2 e^{-3x}$

$$y_p = \frac{\cosh(2x-1) + 3^x}{D^2-4} = \frac{1}{2}\left[\frac{e^{2x-1}}{D^2-4} + \frac{e^{-(2x-1)}}{D^2-4}\right] + \frac{3^x}{D^2-4}$$

$y_p = P_1 + P_2 + P_3$ (say)

$$P_1 = \frac{1}{2} \cdot \frac{e^{2x-1}}{D^2-4} = \frac{1}{2} \cdot \frac{e^{2x-1}}{2^2-4} \quad [\because D_r = 0]$$

$$= \frac{1}{2} x \cdot \frac{e^{2x-1}}{2D}$$

$$= \frac{1}{2} \cdot x \cdot \frac{e^{+(2x-1)}}{+4} = \frac{x}{8} e^{2x-1}$$

$$P_1 = +\frac{x}{8} e^{(2x-1)}$$

$$P_2 = \frac{1}{2} \cdot \frac{e^{-(2x-1)}}{D^2-4} = \frac{1}{2} \cdot \frac{e^{-(2x-1)}}{(-2)^2-4} \quad (D_r = 0]$$

$$P_2 = \frac{1}{2}\cdot x \cdot \frac{e^{-(2x-1)}}{2D}$$

$$P_2 = \frac{1}{2}\cdot x \cdot \frac{e^{-(2x-1)}}{-4}$$

$$\boxed{P_2 = -\frac{x}{8} e^{-(2x-1)}}$$

$$P_3 = \frac{3^x}{D^2-4} = \frac{(e^{\log 3})^x}{D^2-4} = \frac{e^{\log 3 \cdot x}}{D^2-4} = \frac{e^{\log 3 \cdot x}}{(\log 3)^2-4}$$

$$\boxed{P_3 = \frac{3^x}{(\log 3)^2-4}}$$

complete solution :

$y = y_c + y_p$ where

$y_p = P_1 + P_2 + P_3$

$$\therefore y = c_1 e^{2x} + c_2 e^{-2x} + \frac{x}{8} e^{2x-1} - \frac{x}{8} e^{-(2x-1)} + \frac{3^x}{(\log 3)^2-4}$$

Thus, $\boxed{y = c_1 e^{2x} + c_2 e^{-2x} + \frac{x}{4} \sinh(2x-1) + \frac{3^x}{(\log 3)^2-4}}$ //.

(8) Solve : $x'''(t) - 8x(t) = (1-e^t)^2$

$\rightarrow$ we have, $(D^3-8)\,x(t) = (1-e^t)^2$

where $D = \frac{d}{dt}$

A.E is $m^3-8=0$ (or) $(m-2)(m^2+2m+4)=0$

$m=2$; $m^2+2m+4=0$

By solving this eqn.

$$m = \frac{-2 \pm \sqrt{4-16}}{2} = \frac{-2 \pm 2i\sqrt{3}}{2} = -1 \pm i\sqrt{3}$$

$$\therefore x_c = c_1 e^{2t} + e^{-t}(c_2 \cos\sqrt{3}t + c_3 \sin\sqrt{3}t)$$

$$x_p = \frac{(1-e^t)^2}{D^3-8} = \frac{1}{D^3-8} - \frac{2e^t}{D^3-8} + \frac{e^{2t}}{D^3-8}.$$

$$x_p = P_1 + P_2 + P_3 \text{ (say)}$$

$$P_1 = \frac{e^{0t}}{D^3-8} = \frac{e^{0t}}{-8} = \frac{-1}{8}$$

$$P_2 = \frac{-2e^t}{D^3-8} = \frac{-2e^t}{1-8} = \frac{2e^t}{7}$$

$$P_3 = \frac{e^{2t}}{D^3-8} = \frac{e^{2t}}{2^3-8} \cdot (Dr = 0)$$

$$= t \cdot \frac{e^{2t}}{3D^2}$$

$$= t \cdot \frac{e^{2t}}{3 \cdot 2^2} = \frac{te^{2t}}{12}$$

complete soln $x = x_c + x_p$ where $x_p = P_1 + P_2 + P_3$

Thus, $x = c_1 e^{2t} + e^{-t}(c_2 \cos\sqrt{3}t + c_3 \sin\sqrt{3}t) - 1/8 + 2e^t/7 + te^{2t}/12$

* TYPE - 2

P.I of the form $\frac{\sin ax}{f(D^2)}$, $\frac{\cos ax}{f(D^2)}$

$\frac{\sin ax}{f(D^2)} = \frac{\sin ax}{f(-a^2)}$ where $f(-a^2) \neq 0$ [D^2 is replaced by $-a^2$]

The result also holds good for $\frac{\cos ax}{f(D^2)}$

However if $f(-a^2) = 0$,

$\frac{\sin ax}{f(D^2)} = x \cdot \frac{\sin ax}{f'(-a^2)}$; $\frac{\cos ax}{f(D^2)} = x \cdot \frac{\cos ax}{f'(-a^2)}$ where $f'(-a^2) \neq 0$ & so on.

* Note :- ① The results also hold good for the functn $\sin(ax+b)$ & $\cos(ax+b)$

② Function like $\sin ax$, $\cos bx$, $\sin ax \sin bx$, $\cos ax \cos bx$ have to be converted into a sum by using basic trignometric formulae

③ After replacing D^2 by $-a^2$, $f(D)$ transforms & into the form $\alpha D \pm \beta$ (α & β are constants) in which case we x'ly both numerator & denominator by $\alpha D \pm \beta$ for proceeding further in obtaining the P.I.

④ However if $f(D)$ involves only even powers of D then, the same transforms into a constant on replacing D^2 by $-a^2$.

* Example :- ① $\frac{\cos 2x}{D-1}$ we need to x'ly & ÷ by $(D+1)$

$$P.I = \frac{(D+1)\cos 2x}{(D+1)(D-1)} = \frac{D\cos 2x + \cos 2x}{D^2-1} = \frac{-2\sin 2x + \cos 2x}{D^2-1}$$

Here $a = 2$ & we replace D^2 by -4

$$P.I = \frac{-2\sin 2x + \cos 2x}{-4-1} = \frac{1}{5}[2\sin 2x - \cos 2x]$$

Problems

① Solve $y'' = 4y' + 13y = \cos 2x$

→ We have, $(D^2 - 4D + 13)y = \cos 2x$

A.E is, $m^2 - 4m + 13 = 0$ & by solving,

$$m = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \cdot 1 \cdot 13}}{2 \cdot 1} = \frac{4 \pm \sqrt{-36}}{2}$$

$$= \frac{4 \pm \sqrt{36 i^2}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$$

$$\therefore y_c = e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$$

$$y_p = \frac{\cos 2x}{D^2 - 4D + 13}$$

Here, $a = 2$ & hence replace D^2 by $-a^2 = -4$

$$y_p = \frac{\cos 2x}{-4 - 4D + 13} = \frac{\cos 2x}{9 - 4D}$$

Now $\times^{'}$ly & $\div$ by $(9 + 4D)$

$$y_p = \frac{(9 + 4D)\cos 2x}{(9 + 4D)(9 - 4D)} = \frac{9\cos 2x + 4D(\cos 2x)}{81 - 16D^2}$$

$$= \frac{9\cos 2x - 8\sin 2x}{81 - 16(-4)} = \frac{9\cos 2x - 8\sin 2x}{145}$$

Complete soln : $y = y_c + y_p$

Thus, $$\boxed{y = e^{2x}(c_1 \cos 3x + c_2 \sin 3x) + \frac{9\cos 2x - 8\sin 2x}{145}}$$

(2) Solve : $y'' + 9y = \cos 2x \cdot \cos x$

$\rightarrow$ we have, $(D^2+9)y = \cos 2x \cos x$

A.E is $m^2 + 9 = 0 \Rightarrow m = \pm 3i$

$\therefore \quad y_c = c_1 \cos 3x + c_2 \sin 3x$

$$y_p = \frac{1}{2}\frac{(\cos x + \cos 3x)}{D^2+9} = \frac{1}{2}\cdot\frac{\cos x}{D^2+9} + \frac{1}{2}\cdot\frac{\cos 3x}{D^2+9}$$

$y_p = P_1 + P_2$ (say)

* $\sin A \cos B = \frac{1}{2}(\sin(A+B) + \sin(A-B))$
* $\sin A \sin B = \frac{1}{2}[\cos(A+B) - \cos(A-B)]$
* $\cos a \cos b = \frac{\cos(a+b) - \cos(a-b)}{2}$
* $\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$

$$P_1 = \frac{1}{2}\cdot\frac{\cos x}{D^2+9} = \frac{1}{2}\cdot\frac{\cos x}{-1^2+9} = \frac{\cos x}{16}$$

(D^2 is replaced by $-a^2 = -1^2 = -1$)

$$P_2 = \frac{1}{2}\frac{\cos 3x}{D^2+9} = \frac{1}{2}\cdot\frac{\cos 3x}{-3^2+9} \quad (D_r = 0)$$

$$= \frac{1}{2}\, x \cdot \frac{\cos 3x}{2D} = \frac{x}{4}\int \cos 3x\, dx = \frac{x \sin 3x}{12}$$

complete soln :- $y = y_c + y_p$ where $y_p = P_1 + P_2$

Thus, $\boxed{y = c_1 \cos 3x + c_2 \sin 3x + \frac{\cos x}{16} + \frac{x \sin 3x}{12}}$

(3) Solve : $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4\cos^2 x$

$\rightarrow$ we have, $(D^2 + 3D + 2)y = 4\cos^2 x$

A.E is, $m^2 + 3m + 2 = 0$ (or) $(m+1)(m+2) = 0$

$m = -1, -2$

$\therefore \quad y_c = c_1 e^{-x} + c_2 e^{-2x}$

we have, $4\cos^2 x = 2(1 + \cos 2x)$

* $\cos 2x = 2\cos^2 x - 1$
* $\cos^2 x = \frac{1 + \cos 2x}{2}$

$$y_p = \frac{2}{D^2+3D+2} + \frac{2\cos 2x}{D^2+3D+2} = P_1 + P_2 \text{ (say)}$$

$$P_1 = \frac{2e^{0x}}{D^2+3D+2} = \frac{2e^{0x}}{0^2+3\cdot 0+2} = 1$$

$$P_2 = \frac{2\cos 2x}{D^2+3D+2} = 2\left[\frac{\cos 2x}{-4+3D+2}\right] = 2\left[\frac{\cos 2x}{3D-2}\right]$$

$$= \frac{2(3D+2)(\cos 2x)}{(3D+2)(3D-2)} = \frac{2(-6\sin 2x+2\cos 2x)}{9D^2-4}$$

$$= \frac{4(-3\sin 2x+\cos 2x)}{9(-4)-4}$$

$$= \frac{4(-3\sin 2x+\cos 2x)}{-40}$$

$$P_2 = \frac{3\sin 2x - \cos 2x}{10}$$

complete soln : $y = y_c + y_p$ where $y_p = P_1 + P_2$

thus, $\boxed{y = c_1e^{-x} + c_2e^{-2x} + 1 + \frac{3\sin 2x - \cos 2x}{10}}$

(4) Solve : $(D^3-1)y = 3\cos 2x$

$\rightarrow$ A.E is, $(m^3-1) = 0$ (or) $(m-1)(m^2+m+1) = 0$

$m=1$, $m^2+m+1=0$ & by solving this eqn.

$$m = \frac{-1\pm\sqrt{-3}}{2} = \frac{-1\pm i\sqrt{3}}{2} = \frac{-1}{2} \pm \frac{i\sqrt{3}}{2}$$

$$y_c = c_1e^{x} + e^{-x/2}\{c_2\cos(\sqrt{3}x/2) + c_3\sin(\sqrt{3}x/2)\}$$

$$y_p = \frac{3\cos 2x}{D^3-1} = \frac{3\cos 2x}{(D^2\cdot D)-1} \quad \text{now } D^2 \to -2^2 = -4$$

ie, $$y_p = \frac{3\cos 2x}{-4D-1}$$

$$= \frac{3(4D-1)\cos 2x}{-(4D+1)(4D-1)} = \frac{3(-8\sin 2x - \cos 2x)}{-(16D^2-1)}$$

$$y_p = \frac{-3(8\sin 2x + \cos 2x)}{65}$$

complete soln : $y = y_c + y_p$

Thus, $$y = c_1e^x + e^{-x/2}\{c_2\cos(\sqrt{3}x/2) + c_3\sin(\sqrt{3}x/2)\} - \frac{3}{65}(8\sin 2x + \cos 2x)$$

(5). Solve : $\frac{d^3y}{dx^3} + y = 65\cos(2x+1)$

$\rightarrow$

we have $(D^3+1)y = 65\cos(2x+1)$

$$y_c = c_1e^{-x} + e^{x/2}\{c_2\cos(\sqrt{3}/2)x + c_3\sin(\sqrt{3}/2)x\}$$

$$y_p = \frac{65\cos(2x+1)}{D^3+1} = \frac{65\cos(2x+1)}{(D^2\cdot D)+1} \quad \text{now } D^2 \to -4$$

$$y_p = \frac{65\cos(2x+1)}{-4D+1} = \frac{65(1+4D)\cos(2x+1)}{(1+4D)(1-4D)}$$

$$= \frac{65\{\cos(2x+1) - 8\sin(2x+1)\}}{1-16D^2}$$

$$= \frac{65\{\cos(2x+1) - 8\sin(2x+1)\}}{1-16(-4)}$$

$$y_p = \cos(2x+1) - 8\sin(2x+1)$$

Complete solution : $y = y_c + y_p$

Thus, $$\boxed{y = c_1 e^{-x} + e^{x/2}\left\{c_2 \cos(\sqrt{3}/2)x + c_3 \sin(\sqrt{3}/2)x\right\} + \cos(2x+1) - 8\sin(2x+1)}$$ //

* TYPE – 03

P.I of the form $\frac{\phi(x)}{f(D)}$ where $\phi(x)$ is polynomial in x.

we are seeking the particular solⁿ of $f(D)y = \phi(x)$ where

$\phi(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1}x + a_n$

→ It is evident that $y = y_p$ will also be a polynomial in x.

Hence PI is found by division.

→ By writing $\phi(x)$ in descending powers of x and $f(D)$ in ascending powers of D, the division gets completed without any remainder.

→ Quotient so obtained in process of division will be particular integral.

Ex :- (1) $\frac{2x^2 + 4x + 1}{D^2 + 3D + 2}$

$P.I = \frac{2x^2 + 4x + 1}{2 + 3D + D^2}$ for divisⁿ.

$$
\begin{array}{r|l}
 & x^2 - x + 1 \\
\hline
2 + 3D + D^2 & 2x^2 + 4x + 1 \\
 & 2x^2 + 6x + 2 \\
 & (-) \quad (-) \quad (-) \\
\hline
 & -2x - 1 \\
 & -2x - 3 \\
 & (+) \quad (+) \\
\hline
 & +2 \\
 & 2 \\
 & (-) \\
\hline
 & 0
\end{array}
$$

note :- $3D(x^2) = 6x$; $D^2(x^2) = 2$

note :- $3D(-x) = -3$; $D^2(-x) = 0$

$\therefore$ P.I $= x^2 - x + 1$

Problems

① Solve: $y'' + 3y' + 2y = 12x^2$

$\rightarrow$ we have, $(D^2 + 3D + 2)y = 12x^2$

A.E is, $m^2 + 3m + 2 = 0$ (or)

$(m+1)(m+2) = 0 \Rightarrow m = -1, -2$

$y_c = c_1 e^{-x} + c_2 e^{-2x}$

$$y_p = \frac{12x^2}{D^2 + 3D + 2}$$

we need to divide for obtaining the P.I.

$$
\begin{array}{r|l}
 & 6x^2 - 18x + 21 \\
\hline
2 + 3D + D^2 & 12x^2 \\
 & 12x^2 + 36x + 12 \\
\hline
 & -36x - 12 \\
 & -36x - 54 \\
\hline
 & 42 \\
 & 42 \\
\hline
 & 0
\end{array}
$$

$\left[3D(6x^2) = 3\frac{d}{dx}(6x^2) \right.$

$= 36x$

$\left. D^2(6x^2) = 12 \right]$

Hence $y_p = 6x^2 - 18x + 21$

complete solution : $y = y_c + y_p$

Thus, $\boxed{y = c_1 e^{-x} + c_2 e^{-2x} + 6x^2 - 18x + 21}$

② Solve : $(D^3 + 8)y = x^4 + 2x + 1$

$\rightarrow$ A.E is $m^3 + 8 = 0$ (or) $(m+2)(m^2 - 2m + 4) = 0$

$m = -2$ & $m = \frac{2 \pm \sqrt{4 - 16}}{2} = \frac{2 \pm 2i\sqrt{3}}{2} = 1 \pm i\sqrt{3}$

$\therefore y_c = c_1 e^{-2x} + e^{x}\{c_2 \cos(\sqrt{3}x) + c_3 \sin(\sqrt{3}x)\}$

$y_p = \dfrac{x^4+2x+1}{8+D^3}$

PI is found by division.

$$\begin{array}{r|l} & x^4/8 - x/8 + 1/8 \\ \hline 8+D^3 & x^4 + 2x + 1 \\ & x^4 + 3x \\ \hline & -x + 1 \\ & -x - 0 \\ \hline & 1 \\ & 1 \\ \hline & 0 \end{array}$$

$D^3(x^4/8) = 3x$

$\therefore y_p = 1/8(x^4 - x + 1)$

complete soln $y = y_c + y_p$

Thus, $\boxed{y = c_1 e^{-2x} + e^{x}\{c_2 \cos(\sqrt{3}x) + c_3 \sin(\sqrt{3}x)\} + \frac{1}{8}(x^4 - x + 1)}$

(3) solve : $y'' + 2y' + y = 2x + x^2$

$\rightarrow$ we have $(D^2+2D+1)y = 2x + x^2$

A'E is $m^2+2m+1=0$ (or) $(m+1)^2 = 0 \Rightarrow m = -1, -1$

$y_c = (c_1 + c_2 x)e^{-x}$

$y_p = \dfrac{2x+x^2}{D^2+2D+1} = \dfrac{x^2+2x}{1+2D+D^2}$ PI is found by division.

$$\begin{array}{r|l} & x^2 - 2x + 2 \\ \hline 1+2D+D^2 & x^2 + 2x \\ & x^2 + 4x + 2 \\ \hline & -2x + 2 \\ & -2x - 4 \\ \hline & 2 \\ & 2 \\ \hline & 0 \end{array}$$

$\therefore y_p = x^2 - 2x + 2$

complete soln : $y = y_c + y_p$

$\boxed{y = (c_1 + c_2 x)e^{-x} + (x^2 - 2x + 2)}$

(4) Solve : $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = x^2 - 4x - 6$

→ we have $(D^3 + D^2 + 4D + 4)y = x^2 - 4x - 6$

A.E is given by, $m^3 + m^2 + 4m + 4 = 0$

$m^2(m+1) + 4(m+1) = 0$

$(m+1)(m^2+4) = 0$

$\Rightarrow m = -1, m = \pm 2i$

$y_c = c_1 e^{-x} + c_2 \cos 2x + c_3 \sin 2x$

$y_p = \frac{x^2 - 4x - 6}{D^3 + D^2 + 4D + 4}$ P.I is found by division

$$
\begin{array}{l|l}
4 + 4D + D^2 + D^3 & x^2 - 4x - 6 \\
 & x^2 + 2x + 1/2 \\
\hline
 & -6x - (13/2) \\
 & -6x - 6 \\
\hline
 & -1/2 \\
 & -1/2 \\
\hline
 & 0
\end{array}
$$

Hence $y_p = \frac{x^2}{4} - \frac{3x}{2} - 1/8 = 1/8(2x^2 - 12x - 1)$

complete soln : $y = y_c + y_p$

Thus, $y = c_1 e^{-x} + c_2 \cos 2x + c_3 \sin 2x + \frac{1}{8}(2x^2 - 12x - 1)$

* Alternative Method of Type - (3)

→ P.I Given by ; $\psi(x) = \frac{1}{F(D)}(P_m(x))$ — (1)

→ To evaluate right hand side, we write $\frac{1}{F(D)}$ as $[F(D)]^{-1}$ & expand it in ascending powers of D by using binomial expansion. to get expression of form $\frac{1}{F(D)} = [a_0 + a_1 D + a_2 D^2 + \cdots]$

① becomes . $\varphi(x) = \{a_0 + a_1 D + a_2 D^2 + \dots\} (P_m(x))$.

which can simplified by using $D = \frac{d}{dx}$.

* note:-

The following are some standard binomial expansions.

① $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$

② $(1-x)^{-1} = 1 + x + x^2 + x^4 + \dots$

③ $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$

④ $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$

⑤ solve $(D^2 + D - 2)y = 1 + x - x^2$

→ A.E is $(m^2 + m - 2) = 0$

$(m+2)(m-1) = 0$

$m = -2$ & 1 ∴

C.F $y_c = c_1 e^{-2x} + c_2 e^{x}$ — ①

$$P.I = \frac{(1 + x - x^2)}{(D^2 + D - 2)} \quad — ②$$

$$\frac{1}{f(D)} = \frac{1}{(D^2+D-2)} = \frac{1}{(D+2)(D-1)} = \frac{1}{3}\left\{\frac{1}{D-1} - \frac{1}{D+2}\right\}$$

$$= -\frac{1}{3}\left[\frac{1}{1-D} + \frac{1}{(2+D)}\right]$$

$$= -\frac{1}{3}\left[\frac{1}{1-D} + \frac{1}{2}\left[\frac{1}{1+(D/2)}\right]\right]$$

$$= -\frac{1}{3}\left\{(1-D)^{-1} + \frac{1}{2}(1+D/2)^{-1}\right\}$$

$$= -\frac{1}{3}\left\{(1 + D + D^2 + D^3 + \dots) + \frac{1}{2}\left(1 - \frac{D}{2} + \frac{D^2}{4} - \frac{D^3}{8} + \dots\right)\right\}$$

$$= -\frac{1}{3}\left\{\frac{3}{2} + \frac{3}{4}D + \frac{9}{8}D^2 + \dots\right\}$$

eqn ② becomes,

$$P.I = -\frac{1}{3}\left\{\left(\frac{3}{2} + \frac{3}{4}D + \frac{9}{8}D^2 + \dots\right)(1 + x - x^2)\right\}$$

$$= -\frac{1}{3}\left\{\frac{3}{2}(1 + x - x^2) + \frac{3}{4}D(1 + x - x^2) + \frac{9}{8}D^2(1 + x - x^2) + \dots\right\}$$

$= -\frac{1}{3}\left\{\frac{3}{2}(1+x-x^2)+\frac{3}{4}(1-2x)+\frac{9}{8}(-2)+0\right\}$

$= -\frac{1}{3}\left(-\frac{3}{2}x^2\right) = \frac{1}{2}x^2 \quad \text{—— (2)}$

$\therefore$ the general solⁿ of given eqⁿ is

$y = C.F + P.I$

$\boxed{y = c_1 e^{-2x} + c_2 e^{x} + \frac{1}{2}x^2}$ //.

(6) solve $y'' + y' = x^2 + 2x + 4$

$\longrightarrow$

A.E is $m^2 + m = m(m+1) = 0$

$m = 0, \ m = -1$

C.F $y_c = c_1 e^{0x} + c_2 e^{-x} = c_1 + c_2 e^{-x}$.

$P.I = \frac{1}{f(D)}(x^2+2x+4) = \frac{1}{D^2+D}(x^2+2x+4)$

$P.I = \frac{x^2+2x+4}{D(D+1)} = \left(\frac{1}{D} - \frac{1}{D+1}\right)(x^2+2x+4)$

$P.I = \frac{1}{D}(x^2+2x+4) - (1+D)^{-1}(x^2+2x+4)$

$= \int (x^2+2x+4)\,dx - (1-D+D^2 - \cdots)(x^2+2x+4)$

$= \frac{x^3}{3} + x^2 + 4x - \{(x^2+2x+4) - (x^2+2x+4)D$
$+ D^2(x^2+2x+4)\}$

$= \frac{x^3}{3} + x^2 + 4x - \{x^2+2x+4 - (2x+2) + 2\}$

$= \frac{x^3}{3} + \cancel{x^2} + 4x - \cancel{x^2} - \cancel{2x} - 4 + \cancel{2x} + \cancel{2} - \cancel{2}$

$= \frac{1}{3}x^3 + 4x - 4$

$\therefore$ the general solⁿ is $y = C.F + P.I = c_1 + c_2 e^{-x} + \frac{1}{3}x^3 + 4x - 4$

(7) Solve $y'' + 5y' + 6y = x^2 + e^{-2x}$

$\rightarrow$ A.E is $(D^2 + 5D + 6)y = x^2 + e^{-2x}$

A.E $(m^2 + 5m + 6) = 0$

$(m+3)(m+2) = 0$

C.F is $= c_1 e^{-3x} + c_2 e^{-2x}$ — — (1)

P.I is $= \frac{1}{F(D)}(x^2 + e^{-2x}) = \frac{1}{F(D)} x^2 + \frac{1}{F(D)} e^{-2x}$ — (2)

$$\frac{1}{F(D)} x^2 = \frac{x^2}{(D+3)(D+2)} = \left\{\frac{1}{(2+D)} - \frac{1}{(3+D)}\right\} x^2$$

$$= \left\{\frac{1}{2}\left(1 + D/2\right)^{-1} - \frac{1}{3}\left(1 + D/3\right)^{-1}\right\} x^2$$

$$= \left[\frac{1}{2}\left\{1 - D/2 + (D/2)^2 - \cdots\right\} - \frac{1}{3}\left\{1 - D/3 + (D/3)^2 - \cdots\right\}\right] x^2$$

$$= \left[\frac{1}{6} - \frac{5}{36} D + \frac{19}{216} D^2 - \cdots\right] x^2$$

$$= \left[\frac{1}{6} x^2 - \frac{5}{36} D x^2 + \frac{19}{216} D^2 x^2\right]$$

$$= \left[\frac{x^2}{6} - \frac{5}{18} x + \frac{19}{108}\right]$$

Also, $\frac{1}{F(D)} e^{-2x} = x \cdot \frac{1}{F'(D)} e^{-2x} = \frac{x e^{-2x}}{2(-2)+5} = x e^{-2x}$

eqn (2) becomes, $P.I = \frac{1}{6}\left[x^2 - \frac{5}{3} x + \frac{19}{18}\right] + x e^{-2x}$

$\therefore$ general soln is $y = C.F + P.I$

$$y = c_1 e^{-3x} + c_2 e^{-2x} + \frac{1}{6}\left(x^2 - \frac{5}{3} x + \frac{19}{18}\right) + x e^{-2x}$$

(8) solve $y'' - 6y' + 25y = e^{2x} + \sin x + x$

$\longrightarrow$ A.E is $m^2 - 6m + 25 = 0$

$m = 3 \pm 4i$

$\therefore$ C.F is $= e^{3x}(c_1\cos 4x + c_2\sin 4x)$

P.I is $= \frac{1}{F(D)}(e^{2x} + \sin x + x)$

$$P.I = \frac{1}{F(D)}e^{2x} + \frac{1}{F(D)}\sin x + \frac{1}{F(D)}x \quad — (2)$$

consider

$$\frac{e^{2x}}{F(D)} = \frac{1}{F(2)}e^{2x} = \frac{e^{2x}}{2^2 - (6\times 2) + 25} = \frac{e^{2x}}{17}$$

$$\frac{\sin x}{F(D)} = \frac{\sin x}{D^2 - 6D + 25} = \frac{\sin x}{-1^2 - 6D + 25} = -\tfrac{1}{6}\cdot\frac{1}{(D-4)}(\sin x)$$

$$= -\tfrac{1}{6}\cdot\frac{D+4}{D^2 - 16}(\sin x)$$

$$= -\tfrac{1}{6}\frac{(D-4)}{-1^2 - 16}(\sin x)$$

$$= \frac{1}{102}(\cos x + 4\sin x)$$

$$\frac{x}{F(D)} = \frac{1}{D^2 - 6D + 25}(x) = \frac{1}{25}\left\{1 + \frac{D^2}{25} - \frac{6D}{25}\right\}^{-1}(x)$$

$$= \frac{1}{25}\left\{1 - \frac{D^2}{25} - \frac{6D}{25} + \cdots\right\}x$$

$$= \tfrac{1}{25}\left[x + \tfrac{6}{25}\right]$$

(2) becomes

$$P.I = \frac{e^{2x}}{17} + \frac{1}{102}(\cos x + 4\sin x) + \frac{1}{25}\left(x + \tfrac{6}{25}\right)$$

general soln is

$$y = e^{3x}(c_1\cos 4x + c_2\sin 4x) + \frac{1}{17}e^{2x} + \frac{1}{102}(\cos x + 4\sin x) + \frac{1}{25}\left(x + \tfrac{6}{25}\right)$$

* Linear Differential Equations with Variable co-efficients.

We consider methods of solving two particular forms of ordinary linear differential equations of second order with variable co-efficients, known as Cauchy's and Legendre's homogenous linear differential equations.

* CAUCHY'S LINEAR EQUATIONS

The general form of cauchy's homogeneous linear differential eq^n of second-order is

$$x^2 \frac{d^2y}{dx^2} + P_1 x \frac{dy}{dx} + P_2 y = X. \quad \text{--- (1)}$$

where P_1 & P_2 are constants, & X is specified f^n of x.

To solve this eq^n we change the independent variable from x to z by setting

$$z = \log x \quad \text{or} \quad e^z = x \quad \text{--- (2)}$$

Then, we find that

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz} \quad \text{--- (3)}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dz}\right) - \frac{1}{x^2}\frac{dy}{dz}$$

$$= \frac{1}{x}\frac{d}{dz}\left(\frac{dy}{dz}\right)\frac{dz}{dx} - \frac{1}{x^2}\frac{dy}{dz}.$$

$$= \frac{1}{x}\left[\frac{d^2y}{dz^2}\right]\frac{1}{x} - \frac{1}{x^2}\frac{dy}{dz} = \frac{1}{x^2}\left[\frac{d^2y}{dz^2} - \frac{dy}{dz}\right] \quad \text{--- (4)}$$

from ③ & ④ we find that

$$x\frac{dy}{dx} = \frac{dy}{dz} = Dy \quad -⑤$$

$$x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz} = (D^2-D)y = D(D-1)y \quad -⑥$$

Where, now, $D = \frac{d}{dz}$

Using ⑤ & ⑥ the D.E ① becomes

$$\boxed{P_0 D(D-1)y + P_1(Dy) + P_2 y = X}$$

This is a linear differential eqn with constant coefficients which may be solved by using the methods described earlier.

* <u>PROBLEMS</u>

① Solve the differential eqn $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$

→ we put $t = \log x$ so that we have (or) $e^t = x$

$$x\frac{dy}{dx} = Dy, \quad x^2\frac{d^2y}{dx^2} = D(D-1)y$$

Then the given eqn becomes.

$$D(D-1)y + Dy - y = 0 \quad \text{(or)}$$

$$(D^2-1)y = 0$$

∴ A.E is $m^2 - 1 = 0$

roots are $m = \pm 1$

∴ general soln is $y = c_1e^t + c_2e^{-t}$

since $t = \log x$, this becomes.

$y = c_1x + c_2/x$ → This is general soln of given eqn.

(2) Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = (1+x)^2$

→

we have $x^2y'' - 3xy' + 4y = (1+x)^2$ — (1)

put $t = \log x$ (or) $e^t = x$

then we have, $xy' = Dy$,

$x^2 y'' = D(D-1)y$

where $D = \frac{d}{dt}$

Hence (1) becomes, $[D(D-1) - 3D + 4]y = (1+e^t)^2$

$$(D^2 - 4D + 4)y = 1 + 2e^t + e^{2t}$$

A.E is $m^2 - 4m + 4 = 0$

$(m-2)^2 = 0$

$m = 2, 2$

$\therefore$ C.F $y_c = (c_1 + c_2 t)e^{2t}$

P.I $y_p = \frac{1}{D^2-4D+4} + \frac{2e^t}{D^2-4D+4} + \frac{e^{2t}}{D^2-4D+4} = P_1 + P_2 + P_3$ (say)

$P_1 = \frac{e^{0t}}{D^2-4D+4} = \frac{e^{0t}}{0-0+4} = \frac{1}{4}$

$P_2 = \frac{2e^t}{D^2-4D+4} = \frac{2e^t}{1-4+4} = 2e^t$

$P_3 = \frac{e^{2t}}{D^2-4D+4} = \frac{e^{2t}}{4-8+4}$ (Dr = 0)

$P_3 = t \cdot \frac{e^{2t}}{2D-4} = t \cdot \frac{e^{2t}}{4-4}$ (Dr = 0)

$P_3 = t^2 \frac{e^{2t}}{2}$

$\therefore y_p = P_1 + P_2 + P_3$

$$y_p = 1/4 + 2e^t + t^2 \frac{e^{2t}}{2}$$

$\therefore$ complete soln : $y = y_c + y_p$

Thus. $\boxed{y = (c_1 + c_2 \log x) x^2 + \frac{1}{4} + 2x + \frac{x^2 (\log x)^2}{2}}$

(3) Solve : $x \frac{d^2y}{dx^2} - \frac{2y}{x} = x + \frac{1}{x^2}$

$\longrightarrow$ we have : $xy'' - \frac{2y}{x} = x + 1/x^2$ x^{ly} by x we get

$$x^2 y'' - 2y = x^2 + \frac{1}{x} \quad —(1)$$

put $t = \log x$ (or) $e^t = x$

Then we have. $xy' = Dy$

$x^2 y'' = D(D-1)y$ where $D = \frac{d}{dt}$

Hence (1) becomes. $[D(D-1) - 2]y = e^{2t} + e^{-t}$

i.e, $(D^2 - D - 2)y = e^{2t} + e^{-t}$

A.E is $m^2 - m - 2 = 0$

$(m-2)(m+1) = 0$

$m = 2, -1$

$\therefore$ C.F $y_c = c_1 e^{2t} + c_2 e^{-t}$

$$P.I = y_p = \frac{e^{2t}}{D^2 - D - 2} + \frac{e^{-t}}{D^2 - D - 2} = P_1 + P_2 \text{ (say)}$$

$$P_1 = \frac{e^{2t}}{D^2 - D - 2} = \frac{e^{2t}}{4 - 2 - 2} \quad (D_r = 0)$$

$$P_1 = t \cdot \frac{e^{2t}}{2D-1} = t \cdot \frac{e^{2t}}{3} = \frac{te^{2t}}{3}$$

$$P_2 = \frac{e^{-t}}{D^2-D-2} = \frac{e^{-t}}{1+1-2} \quad (Dr=0)$$

$$P_2 = t \cdot \frac{e^{-t}}{2D-1} = t \cdot \frac{e^{-t}}{-3} = \frac{-te^{-t}}{3}$$

$\therefore$ P.I $y_p = P_1 + P_2 = \frac{te^{2t}}{3} - \frac{te^{-t}}{3}$

$\therefore$ complete solution $y = y_c + y_p$

$$y = c_1x^2 + \frac{c_2}{x} + \frac{te^{2t}}{3} - \frac{te^{-t}}{3}$$

$$y = c_1x^2 + \frac{c_2}{x} + \frac{\log x}{3}x^2 - \log x\left(\frac{1}{3x}\right)$$

$$\boxed{y = c_1x^2 + \frac{c_2}{x} + \frac{\log x}{3}\left[x^2 - \frac{1}{x}\right]}$$ //.

(4) Solve : $x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 8y = 65\cos(\log x)$

$\rightarrow$ put $t = \log x$ (or) $x = e^t$. Then we have.

$xy' = Dy$,

$x^2y'' = D(D-1)y$

$x^3y''' = D(D-1)(D-2)y$ where $D = \frac{d}{dt}$

Hence the given DE becomes,

$$[D(D-1)(D-2) + 3D(D-1) + D + 8]y = 65\cos t$$

$$[D^3 - 3D^2 + 2D + 3D^2 - 3D + D + 8]y = 65\cos t$$

(or) $(D^3+8)y = 65\cos t$

A.E is given by, $m^3+8=0$

(or) $(m+2)(m^2-2m+4)=0$

$m=-2$ & $m^2-2m+4=0$

By solving $m^2-2m+4=0$ we have

$$m = \frac{2 \pm \sqrt{4-16}}{2} = \frac{2 \pm 2i\sqrt{3}}{2} = 1 \pm i\sqrt{3}$$

$$\therefore\ y_c = c_1 e^{-2t} + e^{t}\{c_2\cos(\sqrt{3}\cdot t) + c_3\sin(\sqrt{3}\cdot t)\}$$

Also $y_p = \dfrac{65\cos t}{D^3+8}$ we replace D^2 by $-1^2=-1$

$(D^2 \cdot D + 8)$

$$y_p = \frac{65\cos t}{-D+8} = \frac{65(8+D)\cos t}{64-D^2}$$

$$y_p = \frac{65(8\cos t - \sin t)}{65} = 8\cos t - \sin t$$

complete soln : $y = y_c + y_p$

Thus,
$$\boxed{y = \frac{c_1}{x^2} + x\{c_2\cos(\sqrt{3}\log x) + c_3\sin(\sqrt{3}\log x)\} + 8\cos(\log x) - \sin(\log x)}$$

(5) Solve : $x^2y'' + xy' + 9y = 3x^2 + \sin(3\log x)$

$\rightarrow$ put $t = \log x$ (or) $e^t = x$

$xy' = Dy$

$x^2y'' = D(D-1)y$ where $D = \dfrac{d}{dt}$

The given eqn becomes.

$$[D(D-1) + D + 9]y = 3e^{2t} + \sin 3t$$

i.e. $(D^2+9)y = 3e^{2t} + \sin 3t$

A.E is $m^2+9=0$

$m = \pm 3i$

C.F is $y_c = c_1 \cos 3t + c_2 \sin 3t$

$$y_p = \frac{3e^{2t}}{D^2+9} + \frac{\sin 3t}{D^2+9} = P_1 + P_2 \text{ (say)}$$

$$P_1 = \frac{3e^{2t}}{D^2+9} = \frac{3e^{2t}}{2^2+9} = \frac{3e^{2t}}{13}$$

$$P_2 = \frac{\sin 3t}{D^2+9} = \frac{\sin 3t}{-3^2+9} \quad (Dr=0)$$

$$P_2 = t \cdot \frac{\sin 3t}{2D} = t/2 \int \sin 3t \, dt = -\frac{t \cos 3t}{6}$$

complete sol^n $y = y_c + y_p$

$y_p = P_1 + P_2$

Then, $\boxed{y = c_1 \cos(3\log x) + c_2 \sin(3\log x) + \frac{3x^2}{13} - \frac{\log x \cdot \cos(3\log x)}{6}}$

(6) solve: $x^2y'' - xy' + y = \log x$.

$\rightarrow$ put $t = \log x$ (or) $e^t = x$.

the given eq^n becomes,

$$D(D-1)y - Dy + y = t$$

$$(D^2-2D+1)y = t$$

A.E is $m^2-2m+1 = (m-1)^2 = 0$

$\therefore$ C.F is $y_c = (c_1 + tc_2t)e^t$

$\therefore$ C.F $y_c = (c_1 + c_2 \log x)\, x$

$$P.I = \frac{t}{F(D)} = \frac{t}{(D-1)^2}$$

$$= (1-D)^{-2} t$$

$$= [1 + 2D + 3D^2 + \cdots]\, t$$

$$= t + 2$$

P.I $y_p = \log x + 2$

$\therefore$ complete soln. $y = y_c + y_p$

$$y = (c_1 + c_2 \log x)\, x + \log x + 2$$

M.P ① $\longrightarrow$ $x^2 \frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = \sin(2\log x)$

* Legendre's Linear Equation

An eqⁿ of the form

$$a_0 (ax+b)^2 y'' + a_1 (ax+b) y' + a_2 y = \phi(x) \quad \text{--- (1)}$$

is called Legendre's linear eqⁿ of second order.

We use a substitution to reduce the same into a DE with constant coefficients.

put $t = \log(ax+b)$ (or) $e^t = (ax+b)$

$$\frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{dy}{dt} \cdot \frac{a}{ax+b} \quad (\text{or})$$

$$(ax+b)\frac{dy}{dx} = a\frac{dy}{dt}$$

Let $D = \frac{d}{dt}$ so that we have

$$(ax+b)y' = aDy \quad \text{--- (2)}$$

Diff (2) w.r.t x again we get

$$(ax+b)y'' + ay' = a \cdot \frac{d}{dx}\left(\frac{dy}{dt}\right) = a\frac{d}{dt}\left(\frac{dy}{dt}\right)\frac{dt}{dx}$$

$$(ax+b)y'' + ay' = a\frac{d^2y}{dt^2} \cdot \frac{a}{ax+b}$$

$$(ax+b)^2 y'' + a(ax+b)y' = a^2\frac{d^2y}{dt^2}$$

$$(ax+b)^2 y'' + a \cdot aDy = a^2 D^2 y$$

$$(ax+b)^2 y'' = a^2(D^2 - D)y$$

Hence we have, $(ax+b)^2 y'' = a^2 D(D-1)y \quad \text{--- (3)}$

using (2) & (3) in (1) we have,

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$$\left[a_0 \cdot a^2 D(D-1) + a_1 \cdot aD + a_2\right] y = F(t) \quad \text{—— (4)}$$

where the RHS of (1) being $\phi(x) = \phi\left[\frac{e^t - b}{a}\right] = F(t)$

clearly (4) is a linear differential eqn with constant coefficients of the form.

$$f(D) y = F(t) \qquad \text{where } D = \frac{d}{dt}$$

we can solve this eqn as already discussed to obtain the soln y in terms of t which can be converted back to x.

* <u>Note</u> :- we can further show that,

$$(ax+b)^3 y''' = a^3 D(D-1)(D-2) y$$
$$(ax+b)^4 y'''' = a^4 D(D-1)(D-2)(D-3) y \quad \text{etc.}$$

* <u>PROBLEMS</u>

(1) Solve the Legendre's form of linear equation

→ $(1+x)^2 \frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = \sin 2[\log(1+x)]$

→ put $t = \log(1+x)$ (or) $e^t = 1+x$ then we have

$$(1+x)\frac{dy}{dx} = 1 \cdot Dy,$$

$$(1+x^2)\frac{d^2y}{dx^2} = 1^2 D(D-1)y; \qquad \text{where } D = \frac{d}{dt}$$

Hence the given DE becomes,

$$[D(D-1) + D + 1]\, y = \sin 2t$$

i.e $(D^2+1)y = \sin 2t$

A.E is $m^2 + 1 = 0 \Rightarrow m = \pm i$

$\therefore$ CF $y_c = c_1 \cos t + c_2 \sin t$

P.I $y_p = \frac{\sin 2t}{D^2+1} = \frac{\sin 2t}{-2^2+1} = \frac{-\sin 2t}{3}$

complete solution : $y = y_c + y_p$.

Thus $y = c_1 \cos[\log(1+x)] + c_2 \sin[\log(1+x)] - \frac{\sin 2[\log(1+x)]}{3}$

(2) Solve $(2x+1)^2 y'' - 6(2x+1)y' + 16y = 8(2x+1)^2$

$\rightarrow$ put $t = \log(2x+1)$ or $e^t = 2x+1$.

$(2x+1)y' = 2 \cdot Dy$.

$(2x+1)^2 y'' = 2^2 D(D-1)y$ where $D = \frac{d}{dt}$

Hence the given eqn becomes,

$[4D(D-1) - 6 \cdot 2D + 16]y = 8e^{2t}$

$(D^2 - D - 3D + 4)y = 2e^{2t}$, on $\div$ by 4

$(D^2 - 4D + 4)y = 2e^{2t}$

A.E is $m^2 - 4m + 4 = 0$ (or) $(m-2)^2 = 0$

$m = 2, 2$

$\therefore y_c = (c_1 + c_2 t)e^{2t}$

$y_p = \frac{2e^{2t}}{(D-2)^2} = \frac{2e^{2t}}{(2-2)^2}$ $(Dr=0)$

$= t \frac{2e^{2t}}{2(D-2)} = t \frac{e^{2t}}{(2-2)}$ $(Dr=0)$

$y_p = t^2 \frac{e^{2t}}{1} = t^2 e^{2t}$

complete soln $y = [c_1 + c_2 \log(2x+1)](2x+1)^2 + \{\log(2x+1)\}^2 (2x+1)^2$

(3) Solve : $(3x+2)^2y'' + 3(3x+2)y' - 36y = 8x^2+4x+1$

→ put $t = \log(3x+2)$ (or) $e^t = 3x+2$.

$(3x+2)y' = 3 \cdot Dy$

$(3x+2)^2 y'' = 3^2 \cdot D(D-1)y$ where $D = \frac{d}{dt}$

Further $x = \frac{1}{3}(e^t - 2)$. Given eqn becomes.

$[9D(D-1) + 9D - 36]y = 8\,\frac{1}{9}(e^t-2)^2 + 4\,\frac{1}{3}(e^t-2) + 1$

$9(D^2-4)y = 8/9\,(e^{2t} - 4e^t + 4) + 4/3\,(e^t - 2) + 1$

$9(D^2-4)y = 8/9\,e^{2t} - \frac{20}{9}e^t + 17/9$

$(D^2-4)y = 1/81\,(8e^{2t} - 20e^t + 17)$

A.E is $m^2 - 4 = 0 \Rightarrow m = \pm 2$

$\therefore y_c = c_1 e^{2t} + c_2 e^{-2t}$

$y_p = \frac{1}{81}\left[\frac{8e^{2t}}{D^2-4} - \frac{20e^t}{D^2-4} + \frac{17e^{0t}}{D^2-4}\right]$

$= \frac{1}{81}\left[t \cdot \frac{8e^{2t}}{2D} - \frac{20e^t}{-3} + \frac{17}{-4}\right]$

$y_p = \frac{1}{81}\left[t \cdot 2e^{2t} + \frac{20}{3}e^t - 17/4\right]$

complete soln : $y = y_c + y_p$.

Thus $\boxed{y = c_1(3x+2)^2 + \frac{c_2}{(3x+2)^2} + \frac{1}{81}\left[2\log(3x+2)(3x+2)^2 + \frac{20}{3}(3x+2) - 17/4\right]}$

(4) $(1+x^2)\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = \sin\{2\log(1+x)\}$

$\rightarrow$ put $t = \log(1+x)$ (or) $e^t = 1+x$.

$D(D-1)y + Dy + y = \sin 2t$

$(D^2+1) = \sin 2t$

A·E is $\Rightarrow$ $m^2+1 = 0$

$m = \pm 1$

C·F is $c_1 \cos t + c_2 \cos t$

$y_c = c_1 \cos\log(1+x) + c_2 \sin\log(1+x)$

& P·I $= \frac{\sin 2t}{D^2+1}$

$= \frac{\sin 2t}{-2^2+1}$

$= -\frac{1}{3}\sin 2t$

P·I $y_p = -\frac{1}{3}\sin[2\log(1+x)]$

∴ The general soln is

$y = y_c + y_p$

$$\boxed{y = c_1 \cos\log(1+x) + c_2 \sin\log(1+x) - \frac{1}{3}\sin[2\log(1+x)]}$$

(5) Solve: $(1+x^2)y'' + (1+x)y' + y = 2\sin\{\log(1+x)\}$

$\rightarrow$ put $t = \log(1+x)$ (or) $e^t = 1+x$

The given equation becomes.

$D(D-1)y + Dy + y = 2\sin t$

$(D^2+1)y = 2\sin t$

For this eqn, A.E is $m^2+1=0$

$\Rightarrow m = \pm i$

C.F is $y_c = c_1 \cos t + c_2 \sin t$

$y_c = c_1 \cos\log(1+x) + c_2 \sin\log(1+x)$

Also P.I $y_p = \dfrac{2\sin t}{D^2+1}$

$= \dfrac{2}{D^2+1}(\sin t)$

$= 2\left[\dfrac{\sin t}{D^2+1}\right]$

$= 2\left[\dfrac{-t}{2}\cos t\right]$

$y_p = -\log(1+x)\cos\log(1+x)$

$\therefore$ The general soln is

$y = y_c + y_p$

$$\boxed{y = c_1\cos\log(1+x) + c_2\sin\log(1+x) - \log(1+x)\cos\log(1+x)}$$

M.P ② $\rightarrow$ $(2x+3)^2 \dfrac{d^2y}{dx^2} - (2x+3)\dfrac{dy}{dx} - 12y = 6x$

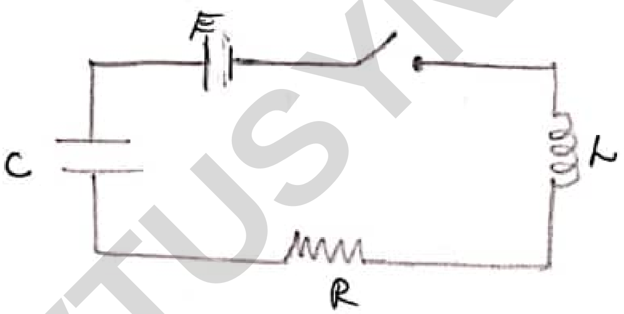
$(2x-1)^2 \dfrac{d^2y}{dx^2} + (2x-1)\dfrac{dy}{dx} - 2y = 8x^2 - 2x + 3$

* Applications of Linear Differential Equations of Second Order

We illustrate the use of linear Differential equations of second-order by considering some examples concerned with electrical circuits & vibrations of a spring.

* Electrical Circuits

consider a simple electric circuit comprised of an inductance of magnitude L (henrys), a resistance of magnitude R (ohms) & capacitance of magnitude C (farads) connected in series. Such a circuit is called an LRC circuit.



LRC circuit

If E is the e.m.f (measured in volts) applied to an LRC circuit then it is known that the current i (measured in amperes) in the circuit at time t is governed by following D.E

$$L\frac{di}{dt} + Ri + q/C = E \quad — (1)$$

Here, q is charge (in coulombs) related to i through the relation $i = dq/dt$.

When expressed in terms of q, eq^n (1) reads

$$L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{C}=E \quad —②$$

when E is specified, the equation ② is to be solved (under the given initial conditions) for determining q in terms of t.

* PROBLEMS

① A particle moves along the x-axis according to the law $\frac{d^2x}{dt^2}+6\frac{dx}{dt}+25x=0$. If the particle is started at $x=0$ with an initial velocity of 12 ft/sec to the left, determine $x(t)$.

→ We have ; $(D^2+6D+25)\,x(t)=0$ &

$x=0$ at $t=0$, $\frac{dx}{dt}=-12$ at $t=0$

[negative sign is due to the movement along the x-axis to the left].

A.E is $m^2+6m+25=0$ & by solving

$$m=\frac{-6\pm\sqrt{36-100}}{2}=\frac{-6\pm\sqrt{64i^2}}{2}=\frac{-6\pm 8i}{2}=-3\pm 4i$$

$$\therefore x=x(t)=e^{-3t}(C_1\cos 4t+C_2\sin 4t) \quad —①$$

Now, $$\frac{dx}{dt}=e^{-3t}(-4C_1\sin 4t+4C_2\cos 4t)-3e^{-3t}(C_1\cos 4t+C_2\sin 4t) \quad —②$$

Using the initial conditions ① & ② respectively becomes,

$0=C_1$ & $-12=4C_2-3C_1$

$C_1=0$ & $C_2=-3$

Thus, $\boxed{x(t)=-3e^{-3t}\sin 4t}$ //

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(2) A voltage $E = E_0 e^{-at}$, where E_0 & a are constants, is applied at time $t>0$ to an LR circuit of inductance L & resistance R. Find the charge and current at time $t>0$, given that the circuit carries no charge & no current at time $t=0$.

→ Here the D.E governing the charge q at time t is

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} = E. \qquad \text{——} (1)$$

It is given that $E = E_0 e^{-at}$ Using this, eq^n (1) becomes.

$$\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} = \frac{E_0}{L}e^{-at}$$

$$\left[D^2 + \frac{R}{L}D\right]q = \frac{E_0}{L}e^{-at} \qquad \text{——} (2)$$

where $D = \frac{d}{dt}$

For this equation, in which t is the independent variable & q is the dependent.

the A.E is $m^2+(R/L)m = 0$ whose roots are
$m(m+R/L)=0$, $m_1=0$. $m+R/L=0 \Rightarrow m=-R/L$
$m_1 = 0$ & $m_2 = -R/L$. therefore.

$$C.F = c_1e^{0t} + c_2e^{-(R/L)t} = c_1 + c_2e^{-(R/L)t} \qquad \text{——} (3)$$

Also, $$P.I = \frac{1}{D^2+(R/L)D}\left[\frac{E_0}{L}e^{-at}\right] = \frac{E_0}{L}\frac{1}{(-a)^2+(R/L)(-a)}e^{-at}$$

$$= \frac{E_0}{aL(a-R/L)}e^{-at}$$

$a^2 - \frac{R}{L}a$
$a(a - R/L)$

∴, the general soln of eq^n (2) is

$$q = C.F + P.I$$

$$= c_1 + c_2 e^{-(R/L)t} + \frac{E_0}{aL(a-R/L)} e^{-at} \quad \text{--- (4)}$$

This gives,

$$i = \frac{dq}{dt} = -\frac{R}{L} c_2 e^{-(R/L)t} - \frac{E_0}{L(a-(R/L))} e^{-at} \quad \text{--- (5)}$$

It is given that $q = 0$ & $i = 0$ at $t = 0$.

Using these initial conditions in (4) & (5), we get

$$0 = c_1 + c_2 + \frac{E_0}{aL(a-R/L)} \quad \text{--- (6)} \qquad \& \qquad 0 = -\frac{R}{L} c_2 - \frac{E_0}{L(a-R/L)} \quad \text{--- (7)}$$

The second of these yields.

$$c_2 = -\frac{E_0}{R(a-R/L)} \quad \text{--- (8)}$$

Then, the 1st condition in (6) yields

$$c_1 = \frac{E_0}{R(a-R/L)} - \frac{E_0}{aL(a-R/L)}$$

$$= \frac{E_0}{(a-R/L)} \left(\frac{1}{R} - \frac{1}{aL} \right] \quad \text{--- (9)}$$

Substituting for c_1 & c_2 ~~from (7) in (8), we get~~

$$q = \frac{E_0}{(a-R/L)} \left(\frac{1}{R} - \frac{1}{aL} \right) - \frac{E_0}{R(a-R/L)} e^{-(R/L)t} + \frac{E_0}{aL(a-R/L)} e^{-at}$$

$$= \frac{E_0}{(a-R/L)} \left[\frac{1}{R} - \frac{1}{aL} - \frac{e^{-(R/L)t}}{R} + \frac{e^{-at}}{aL} \right]$$

$$q = \frac{E_0}{(a-R/L)} \left[\frac{1}{R} \left(1 - e^{-(R/L)t}\right) - \frac{1}{aL} \left(1 - e^{-at}\right) \right] \quad \text{--- (10)}$$

$$\& \quad i = \frac{E_0}{L(a-R/L)} e^{-(R/L)t} - \frac{E_0 \cdot e^{-at}}{L(a-R/L)}$$

$$i = \frac{E_0}{L(a - R/L)}\left(e^{-(R/L)t} - e^{-at}\right) \quad \text{—— (19)}$$

Expressions (18) & (19) give the charge & current respectively at time $t \geq 0$.

(3) An LRC circuit carries an e.m.f of voltage $E = E_0 \sin pt$ where $p^2 = 1/LC$ initially the charge & the current in the circuit are zero, find the charge & the current the circuit at time $t \geq 0$, given that R/L is small.

→ Here, the D.E for the charge q in the circuit is

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = E_0 \sin pt$$

$$\frac{d^2q}{dt^2} + \alpha\frac{dq}{dt} + p^2 q = K \sin pt \quad \text{—— (1)}$$

we have used the given fact that $p^2 = 1/LC$ & put $\alpha = R/L$ & $K = E_0/L$. It is given that α is small.

For the equation (1), the A.E is $m^2 + \alpha m + p^2 = 0$ whose roots are,

$$m = \frac{1}{2}\left\{-\alpha \pm \sqrt{\alpha^2 - 4p^2}\right\} = \frac{1}{2}\left(-\alpha \pm 2i\sqrt{p^2 - \alpha^2/4}\right)$$

$$= -\frac{\alpha}{2} \pm i\beta \quad \text{where } \beta = \sqrt{p^2 - \alpha^2/4}$$

$$\therefore \quad C.F = e^{-(\alpha/2)t}\left[A\cos\beta t + B\sin\beta t\right]$$

Also, with $D = \frac{d}{dt}$, we have.

$$P.I = \frac{1}{D^2 + \alpha D + p^2}(K \sin pt) = \frac{K}{-p^2 + \alpha D + p^2}(\sin pt) = \frac{K}{\alpha}\cdot\frac{1}{D}(\sin pt)$$

$$= \frac{K}{\alpha}\left(-\frac{1}{p}\cos pt\right) = -\frac{E_0}{L}\cdot\frac{L}{R}\frac{1}{p}\cos pt = -\frac{E_0}{Rp}(\cos pt)$$

Accordingly, the general solution for q is

$$q = C.F + P.I$$

$$= e^{-(\alpha/2)t}\left[A\cos\beta t + B\sin\beta t\right) - \frac{E_0}{Rp}(\cos pt) \quad \text{--- (2)}$$

This gives $i = \frac{dq}{dt} = e^{-(\alpha/2)t}\{-A\beta\sin\beta t + B\beta\cos\beta t\}$

$$-\left(\frac{\alpha}{2}\right)e^{-(\alpha/2)t}\{A\cos\beta t + B\sin\beta t\} + \frac{E_0}{R}\sin pt. \quad \text{--- (3)}$$

using the given conditions that $q=0$ & $i=0$ for $t=0$ in (2) & (3), we get.

$$0 = A - \frac{E_0}{Rp} \quad \& \quad 0 = B\beta - \left(\frac{\alpha}{2}\right)A$$

so that

$$A = \frac{E_0}{Rp} \quad \& \quad B = \frac{\alpha}{2\beta}\cdot\frac{E_0}{Rp}$$

substituting these A & B in (2) & (3) we obtain

$$q = \frac{E_0}{Rp}\left[e^{-(\alpha/2)t}\left\{\cos\beta t + \frac{\alpha}{2\beta}\sin\beta t\right\} - \beta\cos pt\right\} \quad \text{--- (4)}$$

$$i = \frac{E_0}{Rp}\left[-e^{-(\alpha/2)t}\left\{\beta + \frac{\alpha^2}{4\beta}\right\}\sin\beta t + p\sin pt\right] \quad \text{--- (5)}$$

These are the expressions for the charge & current at time t.

* <u>Note</u>:- If $\alpha = R/L$ is so small that α^2 & higher powers of α may be ignored, we have

$e^{-(\alpha/2)t} = 1 - \frac{\alpha}{2}t$ & $\beta \approx p$. Then expression (5) becomes

$$i \approx \frac{E_0}{Rp}\left\{-\left(1 - \frac{\alpha}{2}t\right)p\sin pt + p\sin pt\right\}$$

$$= \frac{E_0\alpha t}{2R}\sin pt$$

$$i = \frac{E_0 t}{2L}\sin pt \;//$$

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4) At time $t>0$, an emf of voltage $E = E_0(1-\cos t)$ where E_0 is a constant is applied to an LRC circuit for which $L=R=C=1$. Initially, there is no charge or current in the circuit. Find the charge & the current at time $t>0$.

→ Here the D.E governing the charge q at time t is

$$\frac{d^2q}{dt^2} + \frac{dq}{dt} + q = E_0(1-\cos t).$$

Sol

$$(D^2+D+1)q = E_0(1-\cos t) \quad \text{—— (1)}$$

Where $D = \frac{d}{dt}$

For this eqn, the A.E is $m^2+m+1=0$ whose roots are given by

$$m = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$$\therefore \quad C.F = e^{(-1/2)t}\left\{A\cos\frac{\sqrt{3}}{2}t + B\sin\frac{\sqrt{3}}{2}t\right\}$$

Also,

$$P.I = \frac{1}{D^2+D+1}\left[E_0(1-\cos t)\right]$$

$$= E_0\left\{\frac{1}{D^2+D+1}(e^{0t}) - \frac{1}{D^2+D+1}(\cos t)\right\}$$

$$= E_0\left\{1 - \frac{1}{-1^2+D+1}(\cos t)\right\}$$

$$= E_0\left\{1 - \frac{1}{D}(\cos t)\right\}$$

$$= E_0(1-\sin t)$$

∴ The general soln for charge q is

$$q = C.F + P.I$$

$$= e^{(-1/2)t}\left\{A\cos\frac{\sqrt{3}}{2}t + B\sin\frac{\sqrt{3}}{2}t\right\} + E_0(1-\sin t)$$

This gives the current i as

$$i = \frac{dq}{dt} = e^{(-1/2)t}\left\{ -A\left(\frac{\sqrt{3}}{2}\right)\sin\frac{\sqrt{3}}{2}t + B\left(\frac{\sqrt{3}}{2}\right)\cos\frac{\sqrt{3}}{2}t \right\}$$

$$- \frac{1}{2} e^{(-1/2)t}\left\{ A\cos\frac{\sqrt{3}}{2}t + B\sin\frac{\sqrt{3}}{2}t \right\} - E_0 \cos t.$$

It is given that there is no charge & no current in the circuit at time $t = 0$. Using information in expressions (2) & (3) we get.

$$0 = A + E_0 \quad \& \quad 0 = B\left[\frac{\sqrt{3}}{2}\right] - \frac{1}{2}A - E_0$$

These gives.

$$A = -E_0 \quad \& \quad B = \frac{1}{\sqrt{3}} E_0$$

Putting these into expressions (2) & (3), we get.

$$q = E_0\left\{ e^{(-1/2)t}\left[\frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t - \cos\frac{\sqrt{3}}{2}t\right] + (1 - \sin t)\right\}$$

$$i = E_0\left\{ e^{(-1/2)t}\left[\cos\frac{\sqrt{3}}{2}t + \frac{1}{2}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)\sin\frac{\sqrt{3}}{2}t\right] - \cos t\right\}$$

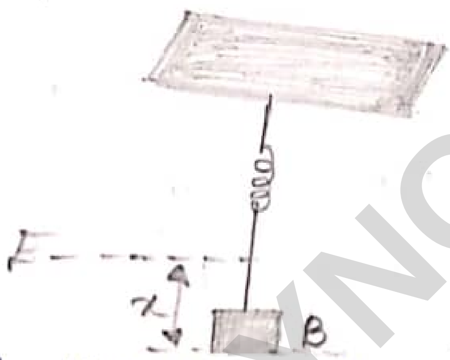
These are the expressions for the charge & the current in the circuit at time $t > 0$.

* Vibration of a Spring

Consider a light elastic spring attached to a support & hanging downward.

Let a body B of small weight be attached to the lower end of spring. When the spring & the body are at rest in an equilibrium positn E, suppose the weight B is pulled downwards and released.

Then, because of elastic property of the spring, the spring vibrates (oscillates) in the vertical direction.



If x denotes the downward displacement of the body B from the equilibrium positn at time t, due to vibration, it can be proved that x satisfies a differential equation of the following form.

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \beta^2 x = f(t) \quad \text{——} (1)$$

Here, $\frac{dx}{dt}$ & $\frac{d^2x}{dt^2}$ respectively represent the velocity & acceleration of the body B at time t which is measured from the instant at which the body is released, $f(t)$ is a time-dependent function associated with extra force, if any, impressed on the body, & β & λ are constants defined by,

$$\beta^2 = \frac{k}{m}, \quad 2\nu = \frac{b}{m},$$

where k is spring constant, m is mass of body B, & b is constant associated with the medium in which the vibration takes place.

Let $x = x_0$ & $\frac{dx}{dt} = v_0$ at $t = 0$ — ②

These are the initial conditions for the vibrations being considered.

The differential equation ① together with the initial conditions ② constitute an initial value problem associated with the vibrations of a spring. The displacement x is determined by solving equation ① under the conditions ②.

* PROBLEMS:

① solve the problem of simple harmonic vibrations of a spring.

→ Here, the governing D.E for vibration is

$$\frac{d^2x}{dt^2} + \beta^2 x = 0 \quad \text{—— ①}$$

initial conditions are

$$x = x_0, \quad \frac{dx}{dt} = v_0 \text{ at } t = 0. \quad \text{—— ②}$$

We note that ① is a homogeneous eqn with t as an independent variable & x as the dependent variable

The A.E is $m^2+\beta^2=0$

whose roots are $\pm\beta i$.

$\therefore$ the general soln of eqn ① is

$$x = A\cos\beta t + B\sin\beta t. \quad \text{—— ③}$$

From this we get. $$\frac{dx}{dt} = -A\beta\sin\beta t + B\beta\cos\beta t. \quad \text{—— ④}$$

When $t=0$, we have $x=x_0$ & $dx/dt = v_0$

See conditions ②. Using these conditions in ① & ②.

we get, $x_0 = A$, $v_0 = B\beta$ (or) $B = v_0/\beta$.

Substituting for A & B from these in ③ we get

$$x = x_0\cos\beta t + \frac{v_0}{\beta}\sin\beta t.$$

This expression gives the displacement x at any time t, & the problem is solved.

② Solve the problem of damped, free vibrations of a spring in the case where the initial velocity is zero.

→ D.E governing vibration is,

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \beta^2 x = 0 \quad \text{—— ①}$$

Initial conditions are

$$x = x_0 \ \& \ \frac{dx}{dt} = 0 \text{ at } t=0 \quad \text{—— ②}$$

eqn ① is homogeneous eqn in which t is independent variable & x is the dependent variable.

A.E is $m^2+2\lambda m+\beta^2=0$ — (3)

roots $\lambda>\beta$, $\lambda=\beta$ (or) $\lambda\le\beta$

we consider these cases one by one.

Case (i): Suppose $\lambda>\beta$ Then the roots of A.E (3) are.

$$m=\frac{-2\lambda\pm\sqrt{4\lambda^2-4\beta^2}}{2}=-\lambda\pm\sqrt{\lambda^2-\beta^2}=\lambda_1,\lambda_2 \text{ say.}$$

Evidently λ_1 & λ_2 are real & distinct.

$\therefore$ the general solution of eqn (1), in this case, is

$$x=c_1e^{\lambda_1 t}+c_2e^{\lambda_2 t} \quad \text{— (4)}$$

from this we get $\frac{dx}{dt}=c_1\lambda_1e^{\lambda_1 t}+c_2\lambda_2e^{\lambda_2 t}$ — (5)

using the initial conditions (2) in (4) & (5), we get

$$x_0=c_1+c_2, \quad 0=c_1\lambda_1+c_2\lambda_2$$

solving these we get,

$$c_1=-\frac{\lambda_2x_0}{\lambda_1-\lambda_2}, \quad c_2=\frac{\lambda_1x_0}{\lambda_1-\lambda_2} \quad \text{— (6)}$$

substituting these into (4) we get.

$$x=\frac{x_0}{\lambda_1-\lambda_2}\left[\lambda_1e^{\lambda_2 t}-\lambda_2e^{\lambda_1 t}\right] \quad \text{— (7)}$$

Case (ii): Suppose $\lambda=\beta$. Then eqn (3) reads

$$m^2+2\lambda m+\lambda^2=0 \quad (\text{or}) \quad (m+\lambda)^2=0$$

for which $-\lambda$ is a repeated root.

∴ in this case, the general soln of eqn (2) as

$$x = (c_1 + c_2 t)e^{-\gamma t} \quad \text{—— (8)}$$

from this we get, $\frac{dx}{dt} = -\gamma(c_1 + c_2 t)e^{-\gamma t} + c_2 e^{-\gamma t}$ —— (9)

using the initial conditions (2) in (8) & (9) we get

$x_0 = c_1$ & $0 = -\gamma c_1 + c_2$ so that $c_2 = \gamma x_0$.

putting these into (8) we get,

$$x = x_0(1 + \gamma t)e^{-\gamma t} \quad \text{—— (10)}$$

case (iii): suppose $\gamma < \beta$. In this case, the roots of eqn (3) are

$$m = \frac{-2\gamma \pm \sqrt{4\gamma^2 - \beta^2}}{2} = -\gamma \pm \alpha i,$$

where $\alpha^2 = \beta^2 - \gamma^2$. Therefore, the general soln of eqn (1), in this case, is

$$x = e^{-\gamma t}(A\cos\alpha t + B\sin\alpha t) \quad \text{—— (11)}$$

from this, we get

$$\frac{dx}{dt} = -\gamma e^{-\gamma t}(A\cos\alpha t + B\sin\alpha t) + e^{-\gamma t}(-A\alpha\sin\alpha t + B\alpha\cos\alpha t) \quad \text{—— (12)}$$

using initial conditions (2) in (11) & (12) we get

$x_0 = A$, & $0 = -\gamma A + B\alpha$ so that $B = \frac{\gamma x_0}{\alpha}$

sub these into (11) we get

$$x = x_0 e^{-\gamma t}\left[\cos\alpha t + \frac{\gamma}{\alpha}\sin\alpha t\right] \quad \text{—— (13)}$$

thus expression (7), (10) & (13) are the general soln of given problem for the cases $\gamma = \beta$, $\gamma > \beta$ & $\gamma < \beta$ respectively.

We note that in the 1st two cases x is always positive & decreases to zero as $t \to \infty$. In the 3rd case, x is oscillatory & x decreases as t increases.

(3) Solve the problem of undamped forced vibrations of a spring governed by the DE $m\frac{d^2y}{dt^2} + ky = f(t)$, in the case where the forcing function is $f(t) = A\sin\omega t$. Take $y(0) = y_0$ & $y'(0) = y_1$

→ We have, $\frac{d^2y}{dt^2} + \frac{k}{m}y = \frac{A}{m}\sin\omega t$

Let, $\lambda^2 = k/m$ & $\mu = A/m$, for convenience,

Now we have, $(D^2+\lambda^2)y(t) = \mu\sin\omega t$

AE is $m^2+\lambda^2 = 0$ $\therefore m = \pm i\lambda$ & $y_c = c_1\cos\lambda t + c_2\sin\lambda t$

$y_p = \frac{\mu\sin\omega t}{D^2+\lambda^2}$, $D^2 \to -\omega^2$ gives $y_p = \frac{\mu\sin\omega t}{\lambda^2-\omega^2}$, $(\lambda \neq \omega)$

$y = y_c + y_p$

Hence, $y(t) = c_1\cos\lambda t + c_2\sin\lambda t + \frac{\mu\sin\omega t}{\lambda^2-\omega^2}$ ——— (1)

Also, $y'(t) = -\lambda c_1\sin\lambda t + \lambda c_2\cos\lambda t + \frac{\mu\omega\cos\omega t}{\lambda^2-\omega^2}$ ——— (2)

Consider $y(0) = y_0$ & $y'(0) = y_1$

(1) & (2) respectively becomes,

$y_0 = c_1$ & $y_1 = \lambda c_2 + \frac{\mu\omega}{\lambda^2-\omega^2}$

$$c_1 = y_0 \ \& \ c_2 = \frac{1}{\lambda}\left[y_1 - \frac{\mu\omega}{\lambda^2-\omega^2}\right]$$

Using these values in ① we have,

$$y(t) = y_0 \cos\lambda t + \frac{1}{\lambda}\left[y_1 - \frac{\mu\omega}{\lambda^2-\omega^2}\right]\sin\lambda t + \frac{\mu\lambda\sin\omega t}{\lambda^2-\omega^2}$$

Thus,

$$\boxed{y(t) = y_0 \cos\lambda t + \frac{y_1}{\lambda}\sin\lambda t + \frac{\mu}{\lambda^2-\omega^2}\left[\sin\omega t - \frac{\omega}{\lambda}\sin\lambda t\right] \quad \lambda \neq \omega}$$

④ Solve the problem of undamped, forced vibrations of a spring in the case where the forcing function is $f(t) = k\sin\omega t$

→ Here, the differential equation for the problem is

$$\frac{d^2x}{dt^2} + \beta^2 x = k\sin\omega t \qquad —①$$

& initial conditions are.

$$x = x_0, \quad \frac{dx}{dt} = v_0 \quad at \quad t = 0 \qquad —②$$

we note that ① is a nonhomogeneous equation with t as an independent variable & x as the dependent variable.

∴ A.E is $m^2+\beta^2 = 0$.

$$C.F = A\cos\beta t + B\sin\beta t.$$

$$P.I = \frac{1}{D^2+\beta^2}(k\sin\omega t) = \frac{k}{(-\omega^2)+\beta^2}(\sin\omega t), \text{ if } \omega \neq \beta.$$

if $\omega = \beta$ we get

$$P.I = \frac{1}{D^2+\beta^2}(k\sin\beta t) = -\frac{kt}{2\beta}\cos\beta t.$$

∵ the general sol^n for x is

$$x = C.F + P.I = \begin{cases} A\cos\beta t + B\sin\beta t + \dfrac{k}{\beta^2-\omega^2}(\sin\omega t) & \text{if } \omega \neq \beta \\ A\cos\beta t + B\sin\beta t - \dfrac{kt}{2\beta}\cos\beta t & \text{if } \omega = \beta \end{cases}$$

from this, we get

$$\frac{dx}{dt} = \begin{cases} -A\beta\sin\beta t + B\beta\cos\beta t + \dfrac{k\omega}{\beta^2-\omega^2}\cos\omega t & \text{if } \omega \neq \beta \\ -A\beta\sin\beta t + B\beta\cos\beta t - \dfrac{k}{2\beta}\{\cos\beta t - t\beta\sin\beta t\} & \text{if } \omega = \beta. \end{cases}$$

Using initial conditions (2) in above expressions, we get

$x_0 = A$ in both the cases, $\omega \neq \beta$, $\omega = \beta$

&

$$v_0 = \begin{cases} B\beta + \dfrac{k\omega}{\beta^2-\omega^2} & \text{if } \omega \neq \beta \\ B\beta - \dfrac{k}{2\beta} & \text{if } \omega = \beta \end{cases}$$

So that

$$B = \begin{cases} \dfrac{1}{\beta}\left(v_0 - \dfrac{k\omega}{\beta^2-\omega^2}\right) & \text{if } \omega \neq \beta \\ \dfrac{1}{\beta}\left(v_0 + \dfrac{k}{2\beta}\right) & \text{if } \omega = \beta \end{cases}$$

Sub for A & B from the above expression in the general $soln$ for x got above,

we obtain.

$$x = \begin{cases} x_0\cos\beta t + \dfrac{v_0}{\beta}\sin\beta t - \dfrac{k}{\beta^2-\omega^2}\left(\dfrac{\omega}{\beta}\sin\beta t - \sin\omega t\right) & \text{if } \omega \neq \beta \\ x_0\cos\beta t + \dfrac{v_0}{\beta}\sin\beta t + \dfrac{k}{2\beta^2}(\sin\beta t - \beta t\cos\beta t) & \text{if } \omega = \beta. \end{cases}$$

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